

Manuscripts considered — urothelial-mets-her2-discordant-kndl

Master inventory: every paper reviewed in this case

Generated 2026-08-21

Manuscripts considered — urothelial-mets-her2-discordant-kndl

Master inventory: 48 clinical (38 included, 10 considered & excluded) + 41 pre-clinical (34 included, 7 considered & excluded) + 39 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Rosenberg/Milowsky (2026) — ESMO Open

Reference: [PMID 42155320](#) · Type: clinical · Inclusion: included

- **Intervention:** enfortumab vedotin plus pembrolizumab — 5-year follow-up
- **Indication / model:** Cisplatin-ineligible, previously untreated locally advanced or metastatic urothelial carcinoma (1L); 84.4% visceral disease; 55.6% with creatinine clearance <60 mL/min. Median follow-up 62.1 months.
- **Design:** Phase 1b/2 dose escalation and cohort A (EV-103/KEYNOTE-869, NCT03288545)
- **n:** 45
- **Effect size:** 73.3 % — confirmed ORR
- **Toxicities:** peripheral sensory neuropathy (25/45, 55.6%); fatigue (23/45, 51.1%); fatigue G_≥3 (5/45, 11.1%); alopecia (22/45, 48.9%); lipase increased G_≥3 (8/45, 17.8%); rash maculo-papular G_≥3 (5/45, 11.1%); treatment-related skin reaction (30/45, 66.7%); serious treatment-related AE (7/45, 15.6%); TRAE leading to dose reduction (15/45, 33.3%); TRAE leading to discontinuation (20/45, 44.4%); treatment-related death G5 (1/45, 2.2%)
- **Case fit:** partial
- **Notes:** The longest published follow-up on EV plus pembrolizumab, with 41.5% alive at five years, which is the reference class for a patient who reached complete response and then stopped for neuropathy. Per-term AE rates come from PMC full text (PMC13213747) because the abstract gives none and ClinicalTrials.gov posts no results for NCT03288545. Grade _≥3 lipase increase at 17.8% is worth weighing against this patient's 2025 pancreatitis.

Kawahara/Nishiyama (2026) — Cancer Rep (Hoboken)

Reference: [PMID 41578675](#) · Type: clinical · Inclusion: included

- **Intervention:** enfortumab vedotin rechallenge after toxicity interruption
- **Indication / model:** Metastatic urothelial carcinoma treated at a single institution, January 2022 to December 2024 (any); 48 patients treated with enfortumab vedotin; 36 developed dermatologic adverse events and 20 resumed the drug after interrupting for them.
- **Design:** Retrospective single-institution observational cohort
- **n:** 48
- **Effect size:** 65.0 % — recurrence of dermatologic adverse events after rechallenge
- **Toxicities:** dermatologic adverse event (36/48, 75%); dermatologic adverse event on rechallenge (13/20, 65%)
- **Case fit:** weak
- **Notes:** The only published cohort addressing enfortumab vedotin rechallenge, and it addresses skin toxicity rather than efficacy or neuropathy. Nothing here speaks to whether re-exposure works after a 25-month response, and neuropathy, the toxicity that stopped this patient, is not the toxicity studied.

Barber-Rotenberg/Tanyi (2026) — Mol Ther

Reference: [PMID 41566776](#) · Type: clinical · Inclusion: included

- **Intervention:** huCAR-meso (fully human anti-mesothelin CAR T cells)
- **Indication / model:** Mesothelin-expressing lung adenocarcinoma, ovarian cancer and mesothelioma (2L+); 14 ovarian, 5 mesothelioma, 1 lung adenocarcinoma. No urothelial patients.
- **Design:** Phase 1 single-arm
- **n:** 20
- **Effect size:** 60.0 % — best overall response
- **Toxicities:** CRS (7/20, 35%)
- **Case fit:** cross_tumor_only
- **Notes:** No objective responses in 20 patients, which is the realistic prior for conventional mesothelin CAR-T in solid tumors regardless of how strong the stain is. Sets the expectation for the mesothelin branch: a validated stain would open trial access, not predict benefit.

Khattak/KEYNOTE-942 Investigators (2026) — J Clin Oncol

Reference: [PMID 42223134](#) · Type: clinical · Inclusion: included

- **Intervention:** intismeran autogene (mRNA-4157/V940) plus pembrolizumab
- **Indication / model:** Resected stage IIIB-IV cutaneous melanoma (adj); 157 randomized 2:1 to combination versus pembrolizumab alone; median planned follow-up 60.3 months at the December 2025 cutoff.
- **Design:** Randomized phase 2b (KEYNOTE-942, NCT03897881)
- **n:** 157
- **Effect size:** 0.51 HR — recurrence-free survival hazard ratio at 5 years
- **Variance:** 95% CI 0.294–0.887
- **Toxicities:** No new safety signals at five years; per-term rates are not reported in this descriptive update.
- **Case fit:** cross_tumor_only
- **Notes:** The most durable randomized signal that adding an individualized neoantigen mRNA vaccine to PD-1 blockade changes outcome, in resected melanoma rather than metastatic urothelial carcinoma. Empty toxicities are a source limit checked three ways: the 5-year update gives a one-sentence safety statement, the paper has no PMC record, and ClinicalTrials.gov posts no results for NCT03897881. The per-term table sits in the 2024 Lancet primary report.

Zhu/Aggarwal (2026) — Clin Cancer Res

Reference: [PMID 41686845](#) · Type: clinical · Inclusion: included

- **Intervention:** ceralasertib (ATR inhibitor) monotherapy in ARID1A-deficient tumors
- **Indication / model:** Locally advanced or metastatic solid tumors with ARID1A loss by immunohistochemistry (2L+); 29 efficacy-evaluable patients, all with radiographic progression at entry and ARID1A loss confirmed by IHC rather than by sequencing alone. 13 had gynecologic malignancies.
- **Design:** Investigator-initiated Simon two-stage phase 2 (NCT03682289)
- **n:** 29
- **Effect size:** 14.0 % — confirmed objective response rate
- **Toxicities:** Safety detail is in the full text rather than the abstract; ceralasertib monotherapy toxicity is dominated by reversible myelosuppression at this schedule.

- **Case fit:** weak
- **Notes:** The only prospective trial that selected patients on ARID1A loss, so it is the ceiling estimate for the ARID1A synthetic-lethality argument in this case, and every response came from gynecologic histology. Toxicities are empty after checking all three sources: the abstract carries no per-term AE data, the paper has no PMC record, and ClinicalTrials.gov posts no results for NCT03682289. Backfill from the Clin Cancer Res safety table when the full text is reachable.

Lee/Makker (2026) — J Gynecol Oncol

Reference:PMID 42135959 · **Type:** clinical · **Inclusion:** excluded — Subgroup report of the endometrial, cervical and ovarian cohorts of the same trial; the bladder-cohort estimand this case needs is in the primary publication.

- **Intervention:** trastuzumab deruxtecan — DESTINY-PanTumor02 gynecologic subgroup
- **Notes:** Excluded: Subgroup report of the endometrial, cervical and ovarian cohorts of the same trial; the bladder-cohort estimand this case needs is in the primary publication. — Read to check whether the exploratory biomarker analyses generalised to the bladder cohort. They are gynecology-specific.

Oh/Meric-Bernstam (2026) — ESMO Open

Reference:PMID 42531802 · **Type:** clinical · **Inclusion:** excluded — Subgroup report covering the biliary tract and pancreatic cohorts, neither of which informs the bladder cohort estimand.

- **Intervention:** trastuzumab deruxtecan — DESTINY-PanTumor02 biliary and pancreatic subgroup
- **Notes:** Excluded: Subgroup report covering the biliary tract and pancreatic cohorts, neither of which informs the bladder cohort estimand.

Vulsteke/KEYNOTE-905/EV-303 Investigators (2026) — N Engl J Med

Reference:PMID 41707170 · **Type:** clinical · **Inclusion:** excluded — Perioperative muscle-invasive bladder cancer; wrong setting for a patient with treated metastatic disease and prior prolonged enfortumab vedotin exposure.

- **Intervention:** perioperative enfortumab vedotin plus pembrolizumab (KEYNOTE-905/EV-303)
- **Notes:** Excluded: Perioperative muscle-invasive bladder cancer; wrong setting for a patient with treated metastatic disease and prior prolonged enfortumab vedotin exposure.

Shen/Zhang (2026) — J Clin Invest

Reference:PMID 41528883 · **Type:** clinical · **Inclusion:** excluded — Preclinical work in human cancer models and directed at R248Q rather than R175H; the researcher agent owns preclinical evidence.

- **Intervention:** TP53 R248Q-directed TCR-engineered T cells
- **Notes:** Excluded: Preclinical work in human cancer models and directed at R248Q rather than R175H; the researcher agent owns preclinical evidence. — Searched deliberately for clinical readouts on the TP53 R175H/HLA-A*02:01 axis and found none: CLSP-1025 (NCT06778863) and NT-175 (NCT05877599) have no published results. The axis is trial-access-only at present.

Hsueh/Tsai (2026) — Ther Adv Med Oncol

Reference:PMID 42422333 · **Type:** clinical · **Inclusion:** excluded — Expression and immune-contexture profiling without treatment outcomes; biomarker-survey material rather than clinical evidence.

- **Intervention:** Nectin-4 and TROP2 expression profiling in urothelial carcinoma
- **Notes:** Excluded: Expression and immune-contexture profiling without treatment outcomes; biomarker-survey material rather than clinical evidence.

Palmieri/Santin (2026) — Gynecol Oncol

Reference:PMID 41690203 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd) in HER2 IHC 1+/2+, FISH-negative tumor
- **Indication / model:** primary FISH-negative endometrial carcinoma cell lines (HER2 IHC 0, 1+, 2+) and HER2 IHC 1+ and 2+ xenografts
- **Design:** HER2 binding at low receptor density still delivers enough DXd to trigger double-strand breaks, supplemented by bystander transfer and ADCC.
- **Effect size:** strong — T-DXd was cytotoxic against IHC 1+ and 2+ FISH-negative lines but not against the IHC 0 line, where killing matched the control ADC. Both IHC 2+ and IHC 1+ xenografts showed growth suppression and longer survival.
- **Variance:** vs non-binding control ADC; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Endometrial models, not urothelial, and the IHC 0 result marks the floor: activity tracked expression tier rather than being independent of it. If repeat validated IHC returns HER2-low rather than 3+, this is the evidence base — weaker and less predictive of benefit than the IHC 3+ data behind the tumor-agnostic approval.

Sethi/Santin (2026) — Gynecol Oncol Rep

Reference:PMID 42500114 · **Type:** preclinical · **Inclusion:** excluded — Activity in this model is driven by activating ERBB2 mutation rather than by expression tier; the patient's ERBB2 mutation status is not in the record, and the case question is whether HER2 protein is retained.

- **Intervention:** trastuzumab deruxtecan in ERBB2-mutated, HER2-low cervical carcinoma
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Activity in this model is driven by activating ERBB2 mutation rather than by expression tier; the patient's ERBB2 mutation status is not in the record, and the case question is whether HER2 protein is retained.

Wang/Ma (2026) — Cancer Cell

Reference:PMID 42167230 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** enfortumab vedotin — uptake-defective resistance with retained NECTIN4
- **Indication / model:** paired pre- and post-treatment spatial transcriptomics of NECTIN4-ADC plus anti-PD-1-treated tumors, humanized mouse models, scRNA-seq, and resistant cell systems
- **Design:** AKR1C1 binds NECTIN4 and blocks AP2M1-dependent clathrin-mediated internalization, while an AKR1C1-WWP2 axis exports the ADC in extracellular vesicles, so payload never accumulates intracellularly.
- **Effect size:** strong — The resistant tumor-cell state carried increased NECTIN4 but defective ADC uptake.

Inhibiting AKR1C1 restored internalization, raised intracellular payload and re-sensitized preclinical models.

- **Variance:** vs pre-treatment paired tumors and parental cell lines; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** No AKR1C1 inhibitor is clinically available. The value here is the second independent line of evidence that post-EV progression can occur with antigen intact, which means a negative functional response to EV does not by itself predict failure of a non-internalizing nectin-4 agent.

Cheville/Shah (2026) — Mod Pathol

Reference:PMID 41173324 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** NECTIN4 amplification and protein expression across urothelial subtypes and nodal metastases
- **Indication / model:** cystectomy tissue microarrays with paired pelvic lymph node metastases; FISH and IHC
- **Design:** 1q23.3 amplification containing NECTIN4 raises protein expression, but copy number and protein are only loosely coupled and both vary by histologic subtype and by site.
- **n:** n=841 patients; n=218 with paired nodal metastasis
- **Effect size:** moderate — NECTIN4 amplification was present in 18% of urothelial carcinomas and 19% of plasmacytoid carcinomas, and amplified tumors carried significantly higher H scores ($P < 0.0001$). Primary-to-node agreement for IHC was only moderate ($\kappa = 0.41$), with amplified primaries occasionally paired to unamplified or low-expressing metastases.
- **Variance:** translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Descriptive pathology with no therapeutic arm, and the cohort is treatment-naïve rather than post-EV. It quantifies the specific risk in this case: a 2023 bladder and liver result does not reliably predict expression in a 2026 para-aortic node, independent of any EV effect.

Weiten/Klümper (2026) — Eur Urol Open Sci

Reference:PMID 41768041 · **Type:** preclinical · **Inclusion:** excluded — Restricted to central nervous system metastases, a site this patient does not have, and published as a letter without a therapeutic arm.

- **Intervention:** NECTIN4 amplification in CNS metastases of urothelial carcinoma
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Restricted to central nervous system metastases, a site this patient does not have, and published as a letter without a therapeutic arm.

Chen/Wang (2026) — Antib Ther

Reference:PMID 42382977 · **Type:** preclinical · **Inclusion:** excluded — Characterizes an investigational epitope-distinct TROP2 ADC (R7059-DXd) with no clinical program this patient could enter; the TROP2-variant binding finding does not change any option on the current list.

- **Intervention:** R7059-DXd (epitope-distinct TROP2 ADC)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Characterizes an investigational epitope-distinct TROP2 ADC (R7059-DXd) with no clinical program this patient could enter; the TROP2-variant binding finding does not change any option on the current

list.

Abrahimi/Merghoub (2026) — J Exp Med

Reference: [PMID 42360231](#) · Type: preclinical · Inclusion: included

- **Intervention:** mesothelin-based CAR T cells in bladder cancer models
- **Indication / model:** multiple bladder cancer cell lines, patient-derived bladder tumor organoids, and xenograft bladder cancer models
- **Design:** A second-generation CAR built on the mesothelin scaffold engages MUC16, the physiologic mesothelin ligand, on bladder tumor cells.
- **Effect size:** strong — MSLN-28z CAR T cells were active across bladder cancer lines and patient-derived organoids, and MUC16 was enriched in tumors refractory to existing therapy. Intravesical delivery gave better tumor control than intravenous transfer while limiting systemic T-cell engraftment.
- **Variance:** vs untransduced T cells; intravenous versus intravesical delivery; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The target engaged is MUC16, not mesothelin itself, and the delivery route was intravesical — designed for organ-confined disease, not a para-aortic node. It is the strongest bladder-specific mesothelin-scaffold data available and it does not transfer directly to systemic mesothelin CAR T dosing.

Scholler/Siegel (2026) — Front Immunol

Reference: [PMID 42459653](#) · Type: preclinical · Inclusion: included

- **Intervention:** juxtamembrane ('stump')-targeting mesothelin CAR T cells
- **Indication / model:** ovarian and pancreatic tumor lines, plus a human mesothelin knock-in mouse model
- **Design:** Most surface mesothelin is proteolytically shed, leaving a juxtamembrane stump; a binder against the stump neither loses its epitope to shedding nor gets decoyed by soluble mesothelin in serum.
- **Effect size:** moderate — CAR 422 T cells matched conventional anti-mesothelin CARs for cytotoxicity and in-vivo activity, and retained activity against tumor cells resistant to the conventional CARs. On-target off-tumor toxicity was lower in the human mesothelin knock-in mouse.
- **Variance:** vs conventional membrane-distal anti-mesothelin CARs (SS1, M5 lineage); translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Ovarian and pancreatic models, no urothelial data, and CAR 422 is not in a trial this patient could enter. It does explain why serum soluble mesothelin is worth measuring before any mesothelin-directed product, and why binder epitope should be read off each protocol.

Takano/Agatsuma (2026) — Mol Cancer Ther

Reference: [PMID 40635151](#) · Type: preclinical · Inclusion: included

- **Intervention:** DS-3939a (TA-MUC1-directed ADC, DXd payload)
- **Indication / model:** clinical tumor tissue microarrays across cancer types, TA-MUC1-positive cell-line-derived and patient-derived xenografts
- **Design:** The antibody reads the aberrantly glycosylated tumor-associated MUC1 glycoform rather than MUC1 protein per se, which is what separates tumor from the ubiquitous normal-tissue expression; the DXd payload

then induces DNA damage and apoptosis.

- **Effect size:** strong — Tissue microarray staining put bladder among the highest TA-MUC1-positive rates, alongside lung and breast. DS-3939a inhibited growth of TA-MUC1-positive cells in vitro and caused regression in cell-line and patient-derived xenografts, including models previously exposed to other cytotoxic ADCs.
- **Variance:** vs isotype control ADC; xenografts pretreated with other cytotoxic ADCs; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Bladder appears in the expression survey but the efficacy models are not urothelial, and DS-3939a is in phase I/II (NCT05875168) rather than on this patient's trial list. The glycoform point is the actionable one: the 2026-08 MXIF read MUC1, and a TA-MUC1 clone would answer a different question.

Jensen/Vincent (2026) — J Immunother Cancer

Reference:PMID 41644270 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRAME425/HLA-A*02:01-directed TCR bispecific engager and TCR T cells
- **Indication / model:** NUT carcinoma cell line panel, patient-derived xenograft, tissue microarray, and targeted immunopeptidomics with labeled peptide standards
- **Design:** A TCR-based bispecific modeled on brenetafusp engages the HLA-A*02:01/SLLQHLIGL (PRAME425) complex and redirects CD3 to lyse the presenting cell.
- **n:** n=5 cell lines; n=77 tissue microarray samples; n=165 sequenced patient samples
- **Effect size:** strong — Targeted mass spectrometry detected the PRAME425 epitope in all four HLA-A*02-positive, PRAME-positive samples tested, at higher levels than the PRAME312 or PRAME462 epitopes. Both the brenetafusp-modeled bispecific and PRAME425 TCR T cells produced T-cell-mediated killing of PRAME-positive cells.
- **Variance:** vs PRAME-negative and HLA-A2-negative targets; BRD4::NUTM1 knockout versus parental; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** NUT carcinoma, a fusion-driven tumor whose PRAME expression is oncogene-driven; nothing here says urothelial carcinoma presents PRAME425 at comparable density. It does show the epitope is directly measurable by mass spectrometry, which is a sharper readout than IHC if PRAME imaging comes back equivocal.

DiNapoli/Vogelstein (2026) — J Clin Invest

Reference:PMID 41542775 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** TP53 R175H/HLA-A*02:01 bispecific antibodies versus CAR T cells — affinity requirements
- **Indication / model:** higher-affinity H2 scFv variants from phage biopanning, tested as bispecific antibodies and as CD28z CAR, CD3γ and TCR formats; mouse xenograft models
- **Design:** Raising scFv affinity for the p53 R175H peptide-HLA complex changes engager potency and receptor function in opposite directions, because a CAR's serial engagement depends on release from a low-density antigen.
- **Effect size:** moderate — Bispecific antibody killing and xenograft tumor control improved as affinity moved into the low nanomolar range, with specificity preserved. The same higher-affinity binders reduced function in every CAR format tested, through weaker T-cell activation on contact.
- **Variance:** vs parental H2 scFv affinity variant across each format; translatability: med

- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Xenografts only. Read alongside the disclosure that the senior group founded Clasp Therapeutics, this is the direct preclinical lineage of the CLSP-1025 engager, and it argues that the engager format — not a CAR — is the right one at this antigen density.

Sodhi/Hartman (2026) — J Immunother Cancer

Reference:PMID 42097835 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** neoantigen vaccine efficacy as a function of antigen localization and CD4 help
- **Indication / model:** syngeneic mouse tumors engineered to express membrane-localized or cytoplasmic neoantigens; pan-cancer TCGA analysis of intratumoral IgG
- **Design:** Membrane-exposed neoantigens elicit class-switched IgG that engages Fc receptors and myeloid cells, a CD4-dependent arm of vaccine efficacy that runs alongside the CD8 response.
- **Effect size:** moderate — Membrane-localized but not cytoplasmic neoantigens drove class-switched IgG, myeloid activation and tumor restriction independent of CD8 T cells, requiring CD4 help, antigen-specific IgG and Fc engagement. A Trop2 T256R mutant with impaired membrane localization failed to elicit comparable vaccine-induced control, and CXCL13 or IL-21 co-expression amplified the antibody arm.
- **Variance:** vs cytoplasmic versus membrane-localized version of the same neoantigen; CD8-depleted and Fc-receptor-deficient arms; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Engineered syngeneic models with defined antigens, far from a 24-peptide personalized program. Relevant because this patient's EliSpot response is CD4-skewed (2 of 14 pools) and currently read as a weak signal — this model gives CD4-dominant responses a plausible effector route, and predicts that where the epitope sits on the protein matters.

Saito/Ono (2026) — Ann Nucl Med

Reference:PMID 41653396 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** [177Lu]Lu-FAPI-46-I (albumin-binder FAP radioligand)
- **Indication / model:** U87MG (high FAP) and HT-1080 (low FAP) tumor-bearing mice; SPECT/CT and radioligand therapy studies
- **Design:** An albumin-binding moiety prolongs blood residence, raising tumor accumulation and dose delivered per administration; killing is by beta emission across the FAP-positive compartment rather than by receptor signaling.
- **Effect size:** moderate — Binding affinity was preserved ($IC_{50} 1.5 \pm 0.3$ nM) while tumor accumulation and retention rose over the parent ligand. The albumin-binder version suppressed and shrank tumors where the parent [177Lu]Lu-FAPI-46 did not.
- **Variance:** vs [177Lu]Lu-FAPI-46 without the albumin binder; vehicle; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Glioblastoma and fibrosarcoma xenografts where FAP is on the tumor cells themselves, not the stromal-FAP configuration seen on this patient's MXIF. The gating problem is upstream anyway: the only FAPi PET on record showed uptake in BCG-granulomatous prostate tissue and no FAP-avid urothelial carcinoma.

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ASP2998 (TROP2-directed STING agonist conjugate), alone or with pembrolizumab, enfortumab vedotin or carboplatin
- **Indication / model:** Locally advanced or metastatic solid tumors; urothelial carcinoma in dose escalation and expansion (2L+); TROP2-directed; prior TROP2 or topoisomerase I-directed therapy explicitly allowed
- **Design:** Phase 1/2 dose escalation and expansion, monotherapy and combination
- **n:** 428
- **Effect size:** — — DLT; treatment-emergent adverse events
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Sacituzumab tirumotecan (sac-TMT, MK-2870) versus paclitaxel, docetaxel or vinflunine
- **Indication / model:** Locally advanced or metastatic urothelial cancer after anti-PD-(L)1, platinum and enfortumab vedotin (2L+); TROP2-directed; no expression cutoff for entry
- **Design:** Randomized open-label phase 3 (TroFuse-031)
- **n:** 590
- **Effect size:** — — Overall survival
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** XYA02 (MUC1-directed agent)
- **Indication / model:** MUC1-expressing advanced, relapsed or refractory solid tumors (2L+); MUC1 expression
- **Design:** Phase 1/2 dose escalation and expansion
- **n:** 190
- **Effect size:** — — Safety, tolerability and preliminary efficacy
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Brenetafusp (IMC-F106C)
- **Indication / model:** Metastatic or unresectable synovial sarcoma and myxoid/round cell liposarcoma in HLA-A

02:01-positive patients (2L+); HLA-A02:01 positive; PRAME expression expected by histology

- **Design:** Phase 2 single-arm in rare PRAME-expressing sarcomas
- **n:** 42
- **Effect size:** — — Objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Rugo/Powell (2025) — Ann Oncol

Reference:PMID 40769277 · **Type:** clinical · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd) — retreatment after grade 1 ILD
- **Indication / model:** HER2-positive, HER2-low or HER2-mutant solid tumors (any); Patients who recovered from a first investigator-graded grade 1, adjudication-confirmed drug-related ILD event and were then retreated (45 of 193 eligible).
- **Design:** Pooled analysis of nine clinical trials
- **n:** 2145
- **Effect size:** 33.3 % — ILD recurrence after retreatment
- **Toxicities:** ILD/pneumonitis recurrence (15/45, 33.3%); ILD/pneumonitis recurrence G1-2 (15/45, 33.3%); ILD/pneumonitis recurrence G $\geq$ 3 (0/45, 0%)
- **Case fit:** weak
- **Notes:** Directly relevant if the 2026-03 CT read is confirmed and T-DXd is held: resuming after a resolved grade 1 event carried no fatal recurrence in this series. Only 45 patients were retreated and they were selected by their treating physicians, so the recurrence rate is a floor rather than an unbiased estimate.

Park/Rugo (2025) — Oncologist

Reference:PMID 41330717 · **Type:** clinical · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd) — gastrointestinal and hematologic AE pooled analysis
- **Indication / model:** Metastatic breast, gastric or lung cancer with varying HER2 expression or HER2 mutation (2L+); Pooled from DS8201-A-J101, DESTINY-Breast01/02/03/04, DESTINY-Lung01/02 and DESTINY-Gastric01/02. Antiemetic analysis limited to before the 2020 protocol change that recommended prophylaxis.
- **Design:** Pooled safety analysis of nine phase 1-3 trials
- **n:** 1678
- **Effect size:** 74.6 % — nausea rate at 5.4 mg/kg
- **Toxicities:** nausea (74.6%); fatigue (56.5%); neutropenia (34.6%); ILD/pneumonitis (adjudicated drug-related) (12%); TEAE leading to dose reduction (22.6%); TEAE leading to discontinuation (17.7%)
- **Case fit:** weak
- **Notes:** Quantifies the toxicity that is currently dose-limiting for this patient: severe nausea after dose 2 sits at the severe end of a very common event, and dose reduction rather than discontinuation is the modal response in the trials. Rates are pooled across breast, gastric and lung studies, not urothelial.

Zhou/Sheng (2025) — Ann Oncol

Reference:PMID 39662628 · **Type:** clinical · **Inclusion:** included

- **Intervention:** disitamab vedotin plus toripalimab
- **Indication / model:** Locally advanced or metastatic urothelial carcinoma, HER2-unselected (1L); 61% treatment-naïve; the rest chemo-refractory. Enrollment was not gated on HER2 expression.
- **Design:** Open-label phase 1b/2 dose escalation and expansion (RC48-C014)
- **n:** 41
- **Effect size:** 73.2 % — confirmed ORR
- **Toxicities:** AST increased (n=41, 65.9%); ALT increased (n=41, 63.4%); ALT increased G_≥3 (n=41, 7.3%); peripheral sensory neuropathy (n=41, 63.4%); gamma-glutamyltransferase increased G_≥3 (n=41, 12.2%); asthenia G_≥3 (n=41, 9.8%); any treatment-related AE G_≥3 (n=41, 51.2%); pneumonitis G5 (1/41, 2.4%)
- **Case fit:** weak
- **Notes:** Shows a HER2 ADC working in urothelial carcinoma without an expression gate, which is the argument that would survive a downgraded re-stain. Neuropathy again reached 63.4% and one patient died of pneumonitis, so both of this case's active toxicity constraints apply.

Powles/Gupta (2025) — Ann Oncol

Reference: [PMID 40460988](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** enfortumab vedotin plus pembrolizumab — 2.5-year update
- **Indication / model:** Previously untreated locally advanced or metastatic urothelial carcinoma (1L); Overall EV-302 population with one additional year of follow-up beyond the primary analysis.
- **Design:** Updated analysis of a randomized phase 3 trial (EV-302, NCT04223856)
- **n:** 886
- **Effect size:** 33.8 months — median overall survival at 2.5-year follow-up
- **Variance:** 95% CI 26.1–39.3
- **Toxicities:** peripheral sensory neuropathy (228/440, 51.8%); peripheral sensory neuropathy (1/440, 0.2%); peripheral motor neuropathy (2/440, 0.5%); rash maculo-papular (143/440, 32.5%); hyperglycaemia (68/440, 15.5%); diarrhoea (161/440, 36.6%); nausea (114/440, 25.9%); pneumonitis (9/440, 2%); ILD/pneumonitis (4/440, 0.9%); pancreatitis (3/440, 0.7%); any serious adverse event (220/440, 50%)
- **Case fit:** strong
- **Notes:** Longer follow-up on the regimen this patient responded to, useful for judging what the 25-month complete response predicts about disease behaviour after stopping. Per-term AE rates come from the ClinicalTrials.gov Adverse Events tab for NCT04223856, which the update itself does not reproduce; the 51.8% sensory-neuropathy rate is the number that explains why this patient stopped at 25 months.

Powles/Grivas (2025) — Ann Oncol

Reference: [PMID 39934055](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** sacituzumab govitecan versus taxane or vinflunine
- **Indication / model:** Advanced urothelial carcinoma after platinum-based chemotherapy and checkpoint inhibitor therapy (2L+); Confirmatory trial for the accelerated approval; median follow-up 9.2 months.
- **Design:** Global open-label randomized phase 3 (TROPiCS-04, NCT04527991)
- **n:** 711
- **Effect size:** 10.3 months — median overall survival
- **Variance:** p=0.087
- **Toxicities:** neutropenia G_≥3 (35%); febrile neutropenia G_≥3 (12%); any treatment-related AE G_≥3 (67%); any treatment-emergent AE G5 (7%)

- **Case fit:** partial
- **Notes:** The negative confirmatory trial behind the October 2024 withdrawal of the urothelial accelerated approval; response rate was higher with sacituzumab govitecan but survival was not. Early neutropenic infection deaths are the mechanism the authors propose for the disconnect, which is a dosing and prophylaxis problem rather than a target problem.

Wermke/Britten (2025) — Nat Med

Reference: [PMID 40205198](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** IMA203 (PRAME-directed autologous TCR-T)
- **Indication / model:** PRAME-positive recurrent or refractory solid tumors in HLA-A*02-positive patients, including melanoma and sarcoma (2L+); 27 patients in phase 1a escalation and 13 in phase 1b extension; 41 started lymphodepletion, 40 received IMA203. PRAME expression was measured by the sponsor's own RT-qPCR assay.
- **Design:** First-in-human open-label 3+3 phase 1 dose escalation and extension (ACTengine, NCT03686124)
- **n:** 40
- **Effect size:** 28.9 % — confirmed objective response rate
- **Toxicities:** CRS G $\geq$ 3 (2/41, 4.9%); neurotoxicity G $\geq$ 3 (0/41, 0%)
- **Case fit:** weak
- **Notes:** The best published clinical evidence for the PRAME/HLA-A*02:01 axis, and the response-versus-expression finding is what makes the pending PRAME result decision-relevant rather than merely eligibility-relevant. Responses were concentrated in melanoma and sarcoma; no urothelial patients were reported. An erratum was published in Nat Med 2025 (10.1038/s41591-025-03731-6).

Saxena/Galsky (2025) — Nat Cancer

Reference: [PMID 40346292](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** PGV001 personalized neoantigen peptide vaccine plus atezolizumab
- **Indication / model:** Urothelial cancer, adjuvant and metastatic settings (any); 12 enrolled; vaccine successfully manufactured for 10 (median 20.3 weeks). Four treated in the adjuvant setting, five in the metastatic setting with measurable disease.
- **Design:** Single-arm phase 1 (NCT03359239)
- **n:** 12
- **Effect size:** on-treatment neoantigen-specific T-cell responses in all treated participants — — feasibility, safety and vaccine-induced neoantigen-specific T-cell immunity
- **Toxicities:** injection-site reaction G1-2 (n=10); fatigue G1-2 (n=10); fever G1-2 (n=10)
- **Case fit:** partial
- **Notes:** The only published personalized neoantigen vaccine trial in urothelial cancer, and the closest published analogue to what this patient is already receiving. Twelve patients, no control arm, and per-term AE percentages are not given in the abstract or the Nature Cancer summary; grade-1 terms are recorded without rates rather than dropped.

Lowery/Rosenberg (2025) — Nat Med

Reference: [PMID 40169866](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** neoantigen-selected tumor-infiltrating lymphocytes, with and without pembrolizumab
- **Indication / model:** Treatment-refractory mismatch-repair-proficient metastatic gastrointestinal cancers (2L+); Three sequential cohorts: 18 patients received bulk unselected TIL, 39 received neoantigen-selected TIL, 34 received neoantigen-selected TIL plus pembrolizumab.
- **Design:** Evolving single-arm phase 2 (NCT01174121, NCI Surgery Branch)
- **n:** 91
- **Effect size:** 23.5 % — objective response rate with neoantigen-selected TIL plus pembrolizumab
- **Variance:** 95% CI 12.4–40.0
- **Toxicities:** severe hematologic toxicity from lymphodepletion G $\geq$ 3 (91/91, 100%); critical care support required G $\geq$ 3 (10%)
- **Case fit:** cross_tumor_only
- **Notes:** This is NCT01174121, the same protocol the trial screener flagged as the neoantigen-TIL route, so the published readout is the direct efficacy prior for that option. Selecting for neoantigen reactivity lifted response from zero to 7.7%, and adding pembrolizumab to 23.5%, at the cost of universal severe cytopenias and a 10% critical-care rate. An erratum was published in Nat Med 2025 (10.1038/s41591-025-03708-5).

Gandara/Graf (2025) — J Immunother Cancer

Reference:PMID 39915003 · **Type:** clinical · **Inclusion:** included

- **Intervention:** tumor mutational burden as a checkpoint-inhibitor biomarker, real-world cohort
- **Indication / model:** 24 advanced cancer types treated with single-agent checkpoint inhibition (any); 8440 patients from a clinico-genomic database drawn from about 280 US treatment facilities, TMB measured by the FDA-approved companion-diagnostic algorithm.
- **Design:** Retrospective real-world cohort with validated mortality endpoint
- **n:** 8440
- **Effect size:** 0.79 HR — real-world overall survival by TMB level
- **Variance:** 95% CI 0.73–0.85
- **Toxicities:** Observational survival analysis; no safety data collected.
- **Case fit:** weak
- **Notes:** Puts this patient's 12.4 mut/Mb in the 10 to <20 band, where the adjusted survival advantage is real but modest, well short of the $\geq$ 20 band. Retrospective, confounded by treatment selection, and no toxicity data by design. An erratum was published in J Immunother Cancer 2025 (10.1136/jitc-2024-010311corr1).

Thomas/Schendel (2025) — Cancers (Basel)

Reference:PMID 41008812 · **Type:** clinical · **Inclusion:** excluded — High-risk myeloid and lymphoid malignancies only; no solid-tumor data to transfer to a urothelial case.

- **Intervention:** MDG1011 PRAME-directed TCR-T
- **Notes:** Excluded: High-risk myeloid and lymphoid malignancies only; no solid-tumor data to transfer to a urothelial case.

Yazdi/Barok (2025) — Cancer Lett

Reference:PMID 40614854 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** HER2-directed ADCs — polyploid giant cancer cell persister resistance

- **Indication / model:** HER2-amplified breast (JIMT-1) and gastric (MKN7, SNU-216) lines; SCID mouse xenografts
- **Design:** ADC exposure selects drug-tolerant polyploid giant cells that downregulate HER2 protein and upregulate nectin-4, moving the vulnerable antigen from one target to another.
- **Effect size:** moderate — All four HER2 ADCs induced polyploid giant cancer cells whose daughter cells carried reduced HER2 protein; JIMT-1 lost the HER2 amplicon after XMT-1522. Those persister populations upregulated nectin-4, and enfortumab vedotin blocked regrowth of JIMT-1 xenografts.
- **Variance:** vs untreated cells and cross-comparison among four HER2 ADCs; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Breast and gastric lines, and the ADC-to-persister sequence is the reverse of this patient's history (EV first, then T-DXd). The morphologic overlap is still hard to ignore: pleomorphic giant cells were described in the 2025 recurrence, and the model predicts exactly the HER2-down/nectin-4-up pattern the 2026-08 MXIF discordance raises.

Chang/Chou (2025) — Nat Commun

Reference:PMID 40931013 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** NECTIN4-directed CAR T cells, with PPAR γ -agonist antigen priming
- **Indication / model:** bladder cancer cell lines across a range of endogenous NECTIN4, EV-resistant derivatives, and a post-EV patient biopsy cohort
- **Design:** PPAR γ transcriptionally controls NECTIN4, so a PPAR γ agonist raises surface antigen; CAR T killing engages antigen directly rather than requiring internalization of a payload.
- **Effect size:** strong — NECTIN4-CAR T cells killed across a range of endogenous NECTIN4 levels, with greater activity against high expressors, and rosiglitazone priming raised NECTIN4 and increased killing. EV-resistant cells remained susceptible, and the post-EV biopsy cohort largely retained NECTIN4 expression.
- **Variance:** vs untransduced T cells; NECTIN4-low versus NECTIN4-high targets; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** No NECTIN4 CAR T product is open to this patient, so the transferable finding is the antigen-retention observation in post-EV human biopsies, which bears on every nectin-4 re-targeting option including the AKY-1189 radioligand and the topoisomerase-payload ADCs.

Jing/Ju (2025) — J Control Release

Reference:PMID 41101697 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Nectin-4 x TROP2 bispecific drug conjugate (dtNDC, eribulin payload)
- **Indication / model:** three human tumor xenograft models (including MDA-MB-468 and HCC1954) in mice
- **Design:** Dual Nectin-4 and TROP2 binding raises the fraction of tumor cells engaged when either antigen alone is patchy, and the Fc-free nanobody format improves tumor penetration.
- **Effect size:** strong — Two subcutaneous doses rendered all MDA-MB-468 tumor-bearing mice tumor-free, and two intravenous doses produced complete remission in HCC1954 xenografts. Liver, lung, kidney and heart histology showed no treatment-related injury at therapeutic doses.
- **Variance:** vs unconjugated and non-fatty-acid-modified formats; vehicle; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only

- **Notes:** Breast xenografts, a nanobody-eribulin conjugate rather than an exatecan ADC, and no arm stratified by low antigen. It supports the dual-antigen concept behind AVZO-103 but not the specific payload or the antigen-low case that matters here.

Santin/Santin (2025) — Cancer Res Commun

Reference:PMID 40862536 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** datopotamab deruxtecan (Dato-DXd) — bystander killing of TROP2-negative tumor
- **Indication / model:** nine primary poorly differentiated endometrial carcinoma cell lines stratified by flow-cytometry TROP2, admixed TROP2-positive/TROP2-negative cocultures, and chemotherapy-resistant xenografts
- **Design:** DXd released inside TROP2-positive cells diffuses into adjacent TROP2-negative cells, adding double-strand breaks and ADCC to direct killing.
- **n:** n=9 primary cell lines
- **Effect size:** strong — Seven of nine primary lines expressed TROP2, and TROP2-positive lines were significantly more sensitive to Dato-DXd than to control ADC. Admixture with TROP2-positive cells produced significant bystander killing of TROP2-negative tumor, and xenograft growth inhibition was significant ($P < 0.0001$).
- **Variance:** vs non-binding control ADC; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Endometrial, not urothelial, and the sensitivity contrast still ran along the expression axis — bystander killing was shown for negative cells adjacent to high expressors, not for a uniformly weak-expressing tumor. Weak TROP2 on research-grade MXIF is therefore a much less reliable predictor of benefit than the strong expression these models carry.

P Shanmugam/Mouw (2025) — Bladder Cancer

Reference:PMID 40034244 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ADC payload sensitivity (MMAE, SN-38) under DNA repair deficiency in bladder cancer
- **Indication / model:** isogenic bladder cancer cell line pairs with and without homologous recombination or nucleotide excision repair deficiency
- **Design:** Topoisomerase I trapping by SN-38 and mitotic damage from MMAE both leave lesions that homologous recombination normally resolves, so HR loss lowers the tolerated payload dose.
- **Effect size:** moderate — BRCA2 loss produced an HR-deficient phenotype and raised sensitivity to cisplatin, PARP inhibition, MMAE and SN-38; NER deficiency did not. SN-38 combined with PARP inhibition was synergistic irrespective of HR or NER status.
- **Variance:** vs repair-proficient isogenic parental lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Isogenic lines only, no in-vivo arm, and ARID1A was not among the repair genes tested — the read-across to this patient's ARID1A loss is inference, not measurement. The retrievable NGS report and somatic HR-gene status would decide whether it applies at all.

Wang/Liu (2025) — Nat Commun

Reference:PMID 40849430 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** M7A-DC / M7GA-DC (mesothelin-directed DARPin-MMAE conjugates)
- **Indication / model:** 3D tumor spheroids and pancreatic cancer mouse models with limited mesothelin expression
- **Design:** A DARPin against the protease-sensitive C-terminal region of mesothelin is internalized and releases MMAE, which then kills adjacent mesothelin-negative cells; the small format penetrates spheroids better than an IgG-based ADC.
- **Effect size:** moderate — Both conjugates were internalized by mesothelin-positive cells and produced bystander killing of mesothelin-negative cells. M7GA-DC improved tumor control and survival in pancreatic models with limited mesothelin, and adding PD-1 blockade generated durable immunologic memory.
- **Variance:** vs conventional full-size mesothelin ADCs; PD-1 blockade alone; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Pancreatic models and a preclinical-only agent. Worth flagging against this patient's constraints rather than for them: the payload is MMAE, the same class that drove his chronic neuropathy on enfortumab vedotin.

Kosti/Dunn (2025) — J Immunother Cancer

Reference:PMID 40081946 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** mesothelin x CD3 T-cell engager (LABC-13F08 BiTE and BiTE-secreting T cells)
- **Indication / model:** MSLN-positive ovarian, mesothelioma and pancreatic cell lines; NSG mice xenografted with the same, using transduced primary human T cells
- **Design:** A bispecific engager crosslinks CD3 on T cells to surface mesothelin; T cells engineered to secrete it locally sustain the engager concentration inside the tumor.
- **Effect size:** moderate — The monovalent BiTE was potent in vitro, and BiTE-secreting T cells controlled ovarian and mesothelioma xenografts with some monotherapy control in a pancreatic model. The secreted engager recruited non-engineered bystander T cells and drove activation-dependent killing of cells lacking mesothelin, and an inducible off-switch cleared the engineered cells.
- **Variance:** vs untransduced primary T cells; monovalent BiTE versus BiTE-secreting T cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** No urothelial model, and the specific construct differs from CT-95. The antigen-independent bystander killing reported here cuts both ways — it may cover heterogeneous expression, and it is also a plausible source of off-tumor toxicity.

Jensen/Vincent (2025) — bioRxiv

Reference:PMID 40161761 · **Type:** preclinical · **Inclusion:** excluded — Preprint of the peer-reviewed publication already logged (PMID 41644270); same data, duplicate citation.

- **Intervention:** PRAME epitopes in NUT carcinoma (bioRxiv preprint)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Preprint of the peer-reviewed publication already logged (PMID 41644270); same data, duplicate citation.

Chen/Chen (2025) — J Immunother Cancer

Reference:PMID 39800373 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** [177Lu]Lu-DOTA-2P(FAPI)2 radioligand therapy with PD-L1 blockade
- **Indication / model:** CT26-FAP syngeneic mouse tumors; immunohistochemistry and single-cell RNA sequencing of the treated microenvironment
- **Design:** FAP-targeted beta emission causes double-strand breaks and raises tumor PD-L1, converting the stromal compartment into an adjuvant for checkpoint blockade.
- **Effect size:** strong — 177Lu-DOTA-2P(FAPI)2 induced double-strand breaks and upregulated PD-L1 more effectively than the 68Ga label. Combined with anti-PD-L1 it eliminated tumors, and rechallenged mice showed complete rejection; single-cell analysis showed more infiltrating CD8 T cells, fewer regulatory T cells, and a contribution from mature neutrophils.
- **Variance:** vs radioligand alone, anti-PD-L1 alone, and untreated mice; 68Ga versus 177Lu label; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** CT26 colorectal cells engineered to express FAP, so the model bypasses the question of whether native stromal FAP delivers enough dose. It offers a mechanistic reason to pair FAP radioligand with checkpoint blockade if imaging ever confirms FAP-avid urothelial disease, and nothing more.

Awwad/Jackson (2025) — Nat Commun

Reference:PMID 39779698 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATR inhibition in ARID1A-deficient cells — KLF5 as a modifier
- **Indication / model:** genome-scale CRISPR screens in ARID1A-deficient and ARID1A-proficient cells under ATR inhibition
- **Design:** KLF5 protects against replication stress partly by regulating BRD4 recruitment to chromatin, so KLF5 loss compounds the replication stress that ARID1A deficiency already imposes.
- **Effect size:** moderate — KLF5 loss severely reduced fitness of ARID1A-deficient cells and sensitized ARID1A-proficient cells to ATR inhibition. The mechanism ran through replication stress, DNA damage, DNA-RNA hybrid accumulation and genomic instability on ATR inhibition.
- **Variance:** vs ARID1A-proficient isogenic cells; untreated screen arm; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Screen-based and in-vitro only. Its practical use in this case is as a caution rather than a lead: ATR-inhibitor response in an ARID1A-mutant tumor is modified by other genes, so a single ARID1A call is a thin basis for predicting benefit.

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** [225Ac]Ac-AKY-1189 (Nectin-4-directed actinium-225 radioligand) with [64Cu]Cu-AKY-1189 imaging
- **Indication / model:** Nectin-4-expressing advanced solid tumors, urothelial carcinoma among the named cohorts (2L+); Nectin-4 expression; [64Cu]Cu-AKY-1189 uptake gates the therapeutic dose
- **Design:** Phase 1 dose escalation and expansion, theranostic pair (64Cu imaging, 225Ac therapy)
- **n:** 150

- **Effect size:** — — DLT and adverse events (Part 1); ORR (Part 2)
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** AVZO-103 (Nectin-4 x TROP2 bispecific ADC, exatecan payload), alone or with a combination agent
- **Indication / model:** Locally advanced or metastatic urothelial cancer and other solid tumors (2L+); Nectin-4 and TROP2 co-targeting; molecular test report required at screening
- **Design:** First-in-human phase 1/2, monotherapy (Part A) and combination (Part B)
- **n:** 355
- **Effect size:** — — DLT and MTD/RP2D (phase 1); ORR (phase 2)
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CLSP-1025 (TP53 R175H peptide-HLA-A*02:01-directed T-cell engager)
- **Indication / model:** Advanced solid tumors harboring TP53 R175H in HLA-A02:01-positive patients; *bladder cancer named among conditions (2L+); TP53 R175H by accredited laboratory test AND HLA-A02:01 by central assay*
- **Design:** Phase 1 monotherapy dose escalation (Part A) and HLA-selected expansion (Part B)
- **n:** 90
- **Effect size:** — — MTD/RDE (Part A); ORR (Part B)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CT-95 (mesothelin x CD3 bispecific antibody)
- **Indication / model:** Advanced cancers associated with mesothelin expression (2L+); Mesothelin expression (tumor types associated with mesothelin expression)
- **Design:** Phase 1a dose escalation and 1b expansion
- **n:** 70
- **Effect size:** — — MTD/recommended dose; treatment-emergent adverse events
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TNhYP218 (anti-mesothelin TNaive/SCM CAR T cells) after fludarabine/cyclophosphamide
- **Indication / model:** Mesothelin-expressing advanced solid tumors including mesothelioma (2L+); Mesothelin expression by on-study screening assay
- **Design:** Phase 1 dose escalation in mesothelin-expressing solid tumors, expansion in mesothelioma
- **n:** 100
- **Effect size:** — — RP2D by DLT; preliminary ORR in mesothelioma
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** E303 (SBE303, Nectin-4-directed antibody-drug conjugate)
- **Indication / model:** Advanced refractory solid tumors (2L+); Nectin-4-directed
- **Design:** Phase 1 first-in-human, all-comer advanced solid tumors
- **n:** 149
- **Effect size:** — — Safety and tolerability; efficacy
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Zanidatamab (HER2-targeted biparatopic bispecific antibody)
- **Indication / model:** HER2-overexpressing (IHC 3+) advanced solid tumors, urothelial carcinoma among the listed conditions (2L+); HER2 IHC 3+ by sponsor-designated central laboratory
- **Design:** Phase 2 single-arm, cohort 1 pan-tumor (HER2-therapy-naive), cohorts 2 and 3 breast and gastroesophageal
- **n:** 200
- **Effect size:** — — Confirmed objective response rate by independent central review
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NEOK002 (EGFR x MUC1 bispecific antibody-drug conjugate)
- **Indication / model:** Select advanced or metastatic solid tumors with no available standard therapy (2L+);

EGFR and MUC1 co-expression (select tumor types)

- **Design:** Phase 1 dose escalation with backfill (part A) and expansion (part B)
- **n:** 155
- **Effect size:** — — Dose-limiting toxicity and recommended dose
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CT-202 (Nectin-4-directed bispecific antibody)
- **Indication / model:** Nectin-4-positive triple-negative breast cancer, colorectal cancer or urothelial cancer (2L+); Nectin-4 positivity
- **Design:** Phase 1a/1b first-in-human dose escalation and expansion
- **n:** 162
- **Effect size:** — — Safety, tolerability and dose finding
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** EB-DT-NK-UC101 (allogeneic cord-blood-derived Nectin-4/HER2 dual-target CAR-NK cells) after lymphodepletion
- **Indication / model:** Advanced or metastatic urothelial carcinoma with Nectin-4 positivity and assessed HER2 status (2L+); Nectin-4 IHC $\geq 1+$ in $\geq 10\%$ of tumor cells with HER2 by IHC/ISH, central review
- **Design:** Phase 1 dose escalation of dual-target allogeneic CAR-NK cells with an inducible caspase-9 safety switch
- **n:** 42
- **Effect size:** — — Safety and tolerability
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** LY4337713 (FAP-directed agent)
- **Indication / model:** FAP-positive pancreatic, breast, ovarian, gastric, colorectal, esophageal and biliary tract cancers (2L+); Clinical or imaging evidence of FAP expression by local assessment
- **Design:** Phase 1 dose escalation and expansion (FiREBOLT)
- **n:** 241
- **Effect size:** — — Safety and dose finding

- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Meric-Bernstam/Lee (2024) — J Clin Oncol

Reference: [PMID 37870536](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd)
- **Indication / model:** HER2-expressing locally advanced or metastatic solid tumors (seven cohorts; bladder cohort n=41), excluding breast, gastric, colorectal and NSCLC (2L+); HER2 IHC 3+ or 2+ by local or central testing, progression after ≥ 1 systemic therapy or no satisfactory alternative; ECOG 0-1. Bladder cohort was metastatic or advanced urothelial carcinoma of renal pelvis, ureter, bladder or urethra. Prior HER2-directed therapy was permitted.
- **Design:** Open-label, multicohort, single-arm phase 2 (DESTINY-PanTumor02, NCT04482309)
- **n:** 267
- **Effect size:** 39.0 % — investigator-assessed confirmed ORR, bladder cohort
- **Variance:** 95% CI 24.2–55.5
- **Toxicities:** any drug-related AE (226/267, 84.6%); any drug-related AE ≥ 3 (109/267, 40.8%); nausea (n=267, 55.1%); vomiting (n=267, 24.7%); diarrhea (n=267, 25.8%); anemia (n=267, 27.7%); fatigue (n=267, 24.7%); ILD/pneumonitis (adjudicated drug-related) (28/267, 10.5%); ILD/pneumonitis (adjudicated drug-related) ≥ 3 (4/267, 1.5%); serious drug-related AE (36/267, 13.5%)
- **Case fit:** partial
- **Notes:** The trial the current T-DXd course rests on; per-term AE rates and the bladder-cohort breakdown were read from PMC full text (PMC10730032) because ClinicalTrials.gov posts no results for NCT04482309. Bladder IHC 3+ rests on 16 patients, so the tier-matched estimate for this patient is wide, and the whole argument depends on whether the current tumor is still 3+ rather than on the 2023-2025 stains.

Bardia/DESTINY-Breast06 Trial Investigators (2024) — N Engl J Med

Reference: [PMID 39282896](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd) — HER2-low/ultralow breast (cross-tumor precedent)
- **Indication / model:** Hormone receptor-positive metastatic breast cancer with HER2-low (IHC 1+ or 2+/ISH-negative) or HER2-ultralow (IHC 0 with membrane staining) expression, after ≥ 1 endocrine-based line and no prior chemotherapy for metastatic disease (2L+); 713 HER2-low and 153 HER2-ultralow patients; primary endpoint PFS by BICR in the HER2-low group.
- **Design:** Randomized open-label phase 3 (DESTINY-Breast06, NCT04494425)
- **n:** 866
- **Effect size:** 13.2 months — median PFS by BICR, HER2-low population
- **Variance:** 95% CI 11.4–15.2; $p < 0.001$
- **Toxicities:** any AE ≥ 3 (n=434, 52.8%); ILD/pneumonitis (adjudicated) (49/434, 11.3%); ILD/pneumonitis (adjudicated) G5 (3/434, 0.7%)
- **Case fit:** cross_tumor_only
- **Notes:** Extends the HER2-low argument down to ultralow staining, again only in breast cancer and only against chemotherapy in a chemotherapy-naïve metastatic population. Overall survival was immature at this analysis, so the ultralow signal rests on PFS alone.

Sheng/Guo (2024) — J Clin Oncol

Reference: [PMID 37988648](#) · Type: clinical · Inclusion: included

- **Intervention:** disitamab vedotin (RC48-ADC)
- **Indication / model:** HER2-positive (IHC 3+ or 2+) locally advanced or metastatic urothelial carcinoma (2L+); Pooled RC48-C005 and RC48-C009; all patients had progressed on ≥ 1 prior systemic chemotherapy line. Subgroups included liver metastases and prior anti-PD-1/L1 exposure.
- **Design:** Pooled analysis of two single-arm phase 2 studies
- **n:** 107
- **Effect size:** 50.5 % — confirmed ORR by blinded independent review
- **Variance:** 95% CI 40.6–60.3
- **Toxicities:** peripheral sensory neuropathy (n=107, 68.2%); peripheral sensory neuropathy ≥ 3 (n=107, 18.7%); leukopenia (n=107, 50.5%); neutropenia (n=107, 42.1%); neutropenia ≥ 3 (n=107, 12.1%); AST increased (n=107, 42.1%); any treatment-related AE ≥ 3 (58/107, 54.2%)
- **Case fit:** partial
- **Notes:** The highest response rate of any HER2-directed agent in urothelial carcinoma, but the payload is MMAE and two-thirds of patients developed sensory neuropathy, which collides with this patient's established CIPN. Chinese-registration data; the drug has no US approval and the registered trials exclude prior HER2-directed therapy.

Sweeney/Meric-Bernstam (2024) — J Clin Oncol

Reference: [PMID 37793085](#) · Type: clinical · Inclusion: included

- **Intervention:** pertuzumab plus trastuzumab
- **Indication / model:** HER2-amplified and/or overexpressed and/or mutated solid tumors, tissue-agnostic cohort (2L+); 263 patients with HER2 amplification and/or overexpression; 83 with HER2 mutation alone. Activity was stratified by KRAS status and by HER2 IHC tier.
- **Design:** Non-randomized phase 2a basket (MyPathway, NCT02091141)
- **n:** 346
- **Effect size:** 25.9 % — ORR in HER2-amplified/overexpressed patients
- **Variance:** 95% CI 20.7–31.6
- **Toxicities:** diarrhoea (129/357, 36.1%); fatigue (105/357, 29.4%); nausea (80/357, 22.4%); anaemia (63/357, 17.6%); infusion-related reaction (51/357, 14.3%); any serious adverse event (95/357, 26.6%); small intestinal obstruction (6/357, 1.7%)
- **Case fit:** partial
- **Notes:** The cleanest published IHC-tier gradient in a pan-tumor HER2 cohort: 41% at 3+ falling to 8.3% at 1+ and zero at IHC 0, which is the quantitative form of the predictive-certainty problem the pending re-stain creates. Per-term AE rates come from the ClinicalTrials.gov Adverse Events tab for NCT02091141 (trastuzumab plus pertuzumab arm, 357 at risk), since the JCO Clinical Trial Update carries only a narrative safety statement.

Powles/EV-302 Trial Investigators (2024) — N Engl J Med

Reference: [PMID 38446675](#) · Type: clinical · Inclusion: included

- **Intervention:** enfortumab vedotin plus pembrolizumab
- **Indication / model:** Previously untreated locally advanced or metastatic urothelial carcinoma (1L);

Cisplatin-eligible and cisplatin-ineligible patients both enrolled; co-primary endpoints PFS by BICR and OS.

- **Design:** Global open-label randomized phase 3 (EV-302/KEYNOTE-A39, NCT04223856)
- **n:** 886
- **Effect size:** 31.5 months — median overall survival
- **Variance:** $p < 0.001$
- **Toxicities:** any treatment-related AE $G \geq 3$ (n=440, 55.9%)
- **Case fit:** strong
- **Notes:** This is the regimen that produced the patient's 25-month complete response, so it anchors what Nectin-4 targeting already achieved for him and sets the bar any re-targeting option is measured against. Per-term neuropathy rates are in the supplement, not the abstract.

Petrylak/Grivas (2024) — J Clin Oncol

Reference: [PMID 39186707](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** sacituzumab govitecan — cisplatin-ineligible, post-checkpoint cohort
- **Indication / model:** Locally advanced or metastatic urothelial carcinoma with progression after a checkpoint inhibitor, in patients cisplatin-ineligible at study entry (2L+); Median age 72.5; 66% visceral metastases (29% liver); half had received platinum only in the perioperative setting.
- **Design:** Single-arm cohort 2 of a multicohort phase 2 study (TROPY-U-01, NCT03547973)
- **n:** 38
- **Effect size:** 32.0 % — ORR by central review
- **Variance:** 95% CI 17.5–48.7
- **Toxicities:** any treatment-emergent AE $G \geq 3$ (n=38, 87%); neutropenia $G \geq 3$ (n=38, 34%); anemia $G \geq 3$ (n=38, 24%); leukopenia $G \geq 3$ (n=38, 19%); fatigue $G \geq 3$ (n=38, 18%); diarrhea $G \geq 3$ (n=38, 16%)
- **Case fit:** partial
- **Notes:** The one published urothelial cohort written for patients who never had cisplatin and progressed after checkpoint therapy, which is close to this patient's actual position. Thirty-eight patients, no randomisation, and 87% grade ≥ 3 toxicity in a population older and frailer than this one.

Loriot/Tagawa (2024) — Clin Cancer Res

Reference: [PMID 39086310](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** sacituzumab govitecan — TROP2 expression versus efficacy
- **Indication / model:** Unresectable locally advanced or metastatic urothelial carcinoma (2L+); Archival tumor from TROPY-U-01 cohorts 1-3; 146 of 192 treated patients had evaluable TROP2 data scored by H-score (0-300) and percentage membrane-positive tumor cells.
- **Design:** Prespecified biomarker analysis within a phase 2 study (TROPY-U-01, NCT03547973)
- **n:** 146
- **Effect size:** no association: benefit observed across all expression levels H-score — association of TROP2 expression with ORR, PFS and OS
- **Toxicities:** Biomarker analysis; safety is reported in the parent cohort publications.
- **Case fit:** partial
- **Notes:** The direct answer to the low-positive TROP2 question: response, PFS and OS were seen across expression levels, so a weak stain does not foreclose a TROP2 ADC. Two limits matter here: 98% of the trial population expressed TROP2 and the median H-score was 215, so the low end of the distribution is thinly represented and this patient's weak multiplex signal sits below almost everyone studied. No toxicities are

recorded because the analysis reports no safety data of its own; the AE profile sits in the TROPY-U-01 cohort rows and ClinicalTrials.gov posts no results for NCT03547973.

Grivas/Loriot (2024) — J Clin Oncol

Reference:PMID 38261969 · **Type:** clinical · **Inclusion:** included

- **Intervention:** sacituzumab govitecan plus pembrolizumab
- **Indication / model:** Metastatic urothelial carcinoma progressing after platinum-based chemotherapy, in checkpoint-inhibitor-naïve patients (2L+); Checkpoint-naïve by design; median age 67, 78% visceral metastases (29% liver). This patient is checkpoint-experienced, which is the mismatch.
- **Design:** Single-arm cohort 3 of a multicohort phase 2 study (TROPY-U-01, NCT03547973)
- **n:** 41
- **Effect size:** 41.0 % — ORR by central review
- **Variance:** 95% CI 26.3–57.9
- **Toxicities:** any treatment-related AE G_≥3 (n=41, 61%); neutropenia G_≥3 (n=41, 37%); leukopenia G_≥3 (n=41, 20%); diarrhea G_≥3 (n=41, 20%)
- **Case fit:** weak
- **Notes:** Adding PD-1 blockade to a TROP2 ADC nearly doubled the single-agent response rate, but only in checkpoint-naïve patients, and this patient progressed on pembrolizumab maintenance. Useful mainly as the ceiling estimate for ADC plus checkpoint combinations rather than as a transferable number.

Lopez/Olive (2024) — Cancer Res Commun

Reference:PMID 39440991 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ETx-22 (nectin-4-directed ADC, exatecan payload)
- **Indication / model:** patient-derived xenografts spanning low to high nectin-4, an MMAE-resistant xenograft, and non-human primate toxicology
- **Design:** An exatecan topoisomerase I payload with a membrane-permeable released species kills at lower antigen occupancy than MMAE and is not a P-glycoprotein substrate to the same degree.
- **Effect size:** strong — ETx-22 produced rapid, durable regressions in PDX models across low to high nectin-4 and in a monomethyl auristatin E-resistant xenograft. Non-human primate tolerability exceeded 20 mg/kg without significant skin toxicity.
- **Variance:** vs MMAE-based enfortumab-vedotin-like ADC; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** PubMed carries only the significance statement; response rates by model and group sizes were read from the PMC full text summary, and no per-arm n is stated there. ETx-22 is not itself on offer — this is class read-across to LY4052031 (camptothecin payload) and AVZO-103 (exatecan payload), and it is the closest preclinical answer to whether a non-MMAE nectin-4 ADC works after EV.

Fukushima/Kobayashi (2024) — Cancer Lett

Reference:PMID 38272345 · **Type:** preclinical · **Inclusion:** excluded — Near-infrared photoimmunotherapy needs light delivered to the lesion, which is not deliverable to a retroperitoneal node and matches no option in this case's trial set.

- **Intervention:** Nectin-4-targeted near-infrared photoimmunotherapy
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Near-infrared photoimmunotherapy needs light delivered to the lesion, which is not deliverable to a retroperitoneal node and matches no option in this case's trial set.

Mog/Zhou (2024) — Sci Transl Med

Reference:PMID 38985855 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Co-STAR receptors against the p53 neoantigen at low antigen density
- **Indication / model:** cancer cells titrated for peptide-HLA density in vitro; mouse tumor models
- **Design:** A receptor combining a TCR-mimic antibody fragment for recognition with MyD88 and CD40 modules for NF-κB costimulation, sitting between a CAR's affinity and a TCR's sensitivity.
- **Effect size:** strong — Co-STAR T cells killed cells bearing low antigen densities better than conventional CARs or patient-derived TCRs. In mice, Co-STARs gave greater T-cell expansion and more durable regressions than TCRs carrying the same MyD88/CD40 costimulation.
- **Variance:** vs conventional CARs and patient-derived TCRs against the same p53 neoantigen; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Preclinical only, and no Co-STAR product is in a trial this patient could enter. It quantifies the shared obstacle for every p53-neoantigen agent — copy number on the cell surface — which is the reason the CLSP-1025 and NT-175 packages should be read as one problem rather than two.

Drakes/DeMuth (2024) — Cancer Immunol Res

Reference:PMID 38270373 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** lymph-node-targeted amphiphile peptide vaccine boosting of TCR T cells
- **Indication / model:** established murine solid tumors refractory to TCR T-cell monotherapy; human TCR T cells against NY-ESO-1, mutant KRAS and HPV16 E7 in vitro
- **Design:** Amphiphile conjugation traffics peptide and adjuvant to draining lymph nodes, where transferred T cells are re-stimulated in situ and the endogenous repertoire is recruited alongside them.
- **Effect size:** strong — Vaccine plus TCR T cells eradicated tumors that TCR T cells alone could not, with expansion of both the transferred cells and the endogenous antitumor repertoire and no lymphodepletion. Durable protection tracked with antigen spreading to tumor antigens the vaccine did not encode.
- **Variance:** vs TCR T cells alone; vaccine alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Murine tumors plus in-vitro human confirmation; no cancer-patient data. It is the mechanistic argument for sequencing this patient's ongoing neoantigen vaccine with a TIL or TCR T product rather than running them as separate programs.

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** LY4052031 (Nectin-4 ADC, camptothecin-98 topoisomerase I payload)

- **Indication / model:** Locally advanced or metastatic urothelial carcinoma and other solid tumors (2L+); Nectin-4-directed; no expression cutoff stated for entry
- **Design:** Phase 1 dose escalation, dose optimization and expansion; cohort B2 restricted to prior-EV patients
- **n:** 420
- **Effect size:** ORR 47% in post-enfortumab-vedotin participants at 3.6 mg/kg (ASCO 2026 initial NEXUS-01 results) — RP2D (phase 1a); ORR (phase 1b)
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** FT825 / ONO-8250 (off-the-shelf iPSC-derived HER2 CAR-T), with or without monoclonal antibodies
- **Indication / model:** Advanced HER2-expressing solid tumors (2L+); HER2 expression (protocol-defined per tumor type)
- **Design:** Phase 1 dose escalation, Regimen A (FT825) and Regimen B (FT825 + cetuximab)
- **n:** 351
- **Effect size:** — — DLT; treatment-emergent adverse events
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CRB-701 (SYS6002, site-specific Nectin-4 ADC with MMAE payload), alone or with anti-PD-1
- **Indication / model:** Advanced or metastatic Nectin-4-expressing solid tumors (2L+); Nectin-4 expression
- **Design:** Phase 1/2 dose escalation, optimization and expansion cohorts
- **n:** 348
- **Effect size:** — — MTD and payload-adjusted dose (Part A); ORR (Parts B and C)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** A2B694 (autologous logic-gated Tmod mesothelin CAR T cells) after lymphodepletion
- **Indication / model:** Mesothelin-expressing solid tumors with tumor loss of heterozygosity of HLA-A*02 (2L+); *Germline HLA-A02 heterozygosity, tumor HLA-A*02 LOH by NGS, and mesothelin expression*
- **Design:** Phase 1/2 of logic-gated CAR T cells, entry through the BASECAMP-1 screening study (NCT04981119)
- **n:** 474
- **Effect size:** — — DLT and RP2D (phase 1); overall response rate (phase 2)
- **Case fit:** weak

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMC-P115C (half-life-extended PRAME-directed ImmTAC)
- **Indication / model:** Advanced PRAME-positive cancers in HLA-A02:01-positive patients (2L+); HLA-A02:01 positive and PRAME-positive per protocol testing
- **Design:** Phase 1 monotherapy dose escalation
- **n:** 140
- **Effect size:** — — Dose-limiting toxicity and adverse events
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Tuvusertib (M1774, ATR inhibitor) with avelumab
- **Indication / model:** Recurrent ARID1A-mutated endometrial cancer (2L+); ARID1A loss-of-function mutation by any CLIA-certified NGS assay
- **Design:** Phase 2 investigator-initiated, single-arm
- **n:** 25
- **Effect size:** — — Objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Kim/Keam (2023) — ESMO Open

Reference: [PMID 37385153](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** trastuzumab (biosimilar trastuzumab-pkrb) plus paclitaxel
- **Indication / model:** HER2-positive recurrent or metastatic urothelial carcinoma (2L+); HER2 positivity by IHC per 2013 ASCO/CAP breast criteria; 26 of 27 enrolled patients were evaluable.
- **Design:** Single-arm phase 2
- **n:** 27
- **Effect size:** 48.1 % — ORR
- **Toxicities:** peripheral neuropathy (n=27, 88.9%); neutropenia G $\geq$ 3 (n=27, 25.9%); thrombocytopenia G $\geq$ 3 (n=27, 7.4%); anemia G $\geq$ 3 (n=27, 7.4%)
- **Case fit:** partial
- **Notes:** A payload-free HER2 antibody paired with a taxane reached 48% response in HER2-positive urothelial carcinoma, but the taxane is what makes it unusable here: neuropathy of any grade hit 88.9% and this patient already carries chronic CIPN. An erratum was published in ESMO Open 2024 (10.1016/j.esmoop.2024.103708).

Hassan/Hong (2023) — Nat Med

Reference:PMID 37501016 · Type: clinical · Inclusion: included

- **Intervention:** gavocabtagene autoleucel (gavo-cel, mesothelin-directed TCR fusion construct T cells)
- **Indication / model:** Treatment-refractory mesothelin-expressing solid tumors (mesothelioma, ovarian, cholangiocarcinoma) (2L+); 32 patients dosed, 29 after lymphodepletion; 30 evaluable for response. Urothelial carcinoma was not among the enrolled histologies.
- **Design:** Phase 1 dose escalation within an ongoing phase 1/2 trial (NCT03907852)
- **n:** 32
- **Effect size:** 20.0 % — overall response rate
- **Toxicities:** pneumonitis G $\geq$ 3 (n=32, 16%); CRS G $\geq$ 3 (n=32, 25%); bronchioalveolar hemorrhage G5 (1/32)
- **Case fit:** cross_tumor_only
- **Notes:** The strongest clinical activity signal for mesothelin-directed cell therapy so far, and the authors call the therapeutic window narrow. Pulmonary toxicity is the dose-limiting event, which matters given the unresolved chest CT question in this case.

Rojas/Balachandran (2023) — Nature

Reference:PMID 37165196 · Type: clinical · Inclusion: included

- **Intervention:** autogene cevumeran (individualized uridine mRNA-lipoplex neoantigen vaccine) with atezolizumab and mFOLFIRINOX
- **Indication / model:** Resected pancreatic ductal adenocarcinoma, adjuvant setting (adj); 16 patients received atezolizumab and vaccine; 15 went on to mFOLFIRINOX. Vaccines were manufactured in real time from the surgical specimen.
- **Design:** Single-arm phase 1
- **n:** 16
- **Effect size:** 50.0 % — vaccine-induced neoantigen-specific T-cell responses and recurrence-free survival by immune response
- **Variance:** p=0.003
- **Toxicities:** any AE attributable to the vaccine G $\geq$ 3 (1/16, 6%); any AE attributable to the vaccine G1-2 (16/16, 100%); any AE attributable to atezolizumab G $\geq$ 3 (0/19, 0%)
- **Case fit:** cross_tumor_only
- **Notes:** The template for reading this patient's EliSpot data: immunological response, not vaccination itself, tracked with outcome, and only half of vaccinated patients mounted one. His 2 of 14 evaluable pools with CD4-skewed reactivity sits at the weak end of that distribution. Grade-level AE data come from PMC full text (PMC10171177); no results are posted on ClinicalTrials.gov and the paper reports attribution by drug rather than per CTCAE term.

Grimm/Zengerling (2023) — Lancet Oncol

Reference:PMID 36868252 · Type: clinical · Inclusion: included

- **Intervention:** nivolumab induction with high-dose ipilimumab boost
- **Indication / model:** Metastatic or unresectable urothelial cancer of the bladder, urethra, ureter or renal pelvis after platinum-based chemotherapy (2L+); 83 patients, Karnofsky $\geq$ 70, measurable disease; 50 (60%) received at least one ipilimumab boost, given either for early non-response or for later progression on nivolumab

maintenance.

- **Design:** Multicentre single-arm phase 2 (TITAN-TCC, NCT03219775)
- **n:** 83
- **Effect size:** 33.0 % — confirmed investigator-assessed ORR, intention-to-treat
- **Variance:** 95% CI 24.0–42.0; $p=0.0049$
- **Toxicities:** immune-mediated enterocolitis G3-4 (9/83, 11%); diarrhea G3-4 (5/83, 6%); treatment-related death G5 (2/83, 2%)
- **Case fit:** partial
- **Notes:** The registered trial of exactly the manoeuvre the care team is considering, adding high-dose ipilimumab after PD-1 failure in urothelial carcinoma, and it produced responses in patients who had already failed nivolumab. Two treatment-related deaths from enterocolitis in 83 patients is the number to weigh against this patient's 2025 checkpoint-associated pancreatitis; entry also required prior platinum.

Rosenberg/Petrylak (2023) — Ann Oncol

Reference: [PMID 37678672](#) · **Type:** clinical · **Inclusion:** excluded — Same cohort as the EV-301 primary publication already included; longer follow-up does not change the post-platinum population mismatch that limits its relevance here.

- **Intervention:** enfortumab vedotin — EV-301 24-month follow-up
- **Notes:** Excluded: Same cohort as the EV-301 primary publication already included; longer follow-up does not change the post-platinum population mismatch that limits its relevance here.

Yu/Wang (2023) — Cancer Res

Reference: [PMID 37306706](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PARP inhibition with temozolomide in ARID1A-mutated cancers
- **Indication / model:** ARID1A-deficient and ARID1A-proficient cancer cells; ovarian tumor xenografts carrying ARID1A mutations
- **Design:** ARID1A deficiency leaves accumulated DNA base lesions and abasic sites with delayed long-patch base excision repair, so adding a methylating agent to PARP trapping converts unrepaired lesions into double-strand breaks.
- **Effect size:** moderate — ARID1A-deficient cells accumulated base lesions and abasic sites and recruited long-patch BER effectors slowly. Temozolomide alone was inactive, but combined with a PARP inhibitor it produced double-strand breaks, replication stress and fork instability, and significantly delayed growth of ARID1A-mutant ovarian xenografts.
- **Variance:** vs temozolomide alone, PARP inhibitor alone, ARID1A-proficient cells; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Ovarian models and a combination with no urothelial trial behind it. The negative half is the useful half: PARP inhibitor monotherapy was not the active arm, which argues against reading ARID1A loss as a PARP-inhibitor indication on its own.

Zhou/Mouw (2023) — Sci Adv

Reference: [PMID 37992168](#) · **Type:** preclinical · **Inclusion:** excluded — Bladder-specific and mechanistically adjacent, but the biomarker tested is ATM loss rather than ARID1A; the patient's ATM status is not in the available

record, so it supports no ranked option.

- **Intervention:** ATM deficiency and DNA repair-targeted vulnerabilities in bladder cancer
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Bladder-specific and mechanistically adjacent, but the biomarker tested is ATM loss rather than ARID1A; the patient's ATM status is not in the available record, so it supports no ranked option.

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** NT-175 (autologous TCR-T targeting TP53 R175H) after fludarabine/cyclophosphamide lymphodepletion
- **Indication / model:** Advanced solid tumors positive for HLA-A02:01 and *TP53 R175H* (*any solid tumor histology accepted*) (2L+); *TP53 R175H variant AND HLA-A02:01* (at least one allele)
- **Design:** Phase 1, single infusion of engineered T cells with post-infusion IL-2
- **n:** 45
- **Effect size:** — — Safety and DLT; preliminary antitumor activity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** LCB84 (TROP2-directed ADC), alone or with an anti-PD-1 antibody
- **Indication / model:** Advanced solid tumors refractory to standard therapy (2L+); TROP2-directed; prior TROP2-directed therapy permitted
- **Design:** Phase 1 dose escalation followed by phase 2 expansion cohorts
- **n:** 300
- **Effect size:** — — Safety and dose finding; objective response rate
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** DS-3939a (TA-MUC1-directed antibody-drug conjugate, DXd-class topoisomerase I payload)
- **Indication / model:** Advanced or metastatic solid tumors (all-comer dose escalation; protocol-defined expansion cohorts) (2L+); TA-MUC1 (tumor-associated MUC1 glycoform)-directed; dose escalation not gated on a MUC1 assay, expansion requires baseline tumor tissue for protocol biomarker work
- **Design:** First-in-human phase 1/2, dose escalation (Part 1) and multi-cohort dose expansion (Part 2)
- **n:** 590
- **Effect size:** — — DLT and adverse events (Part 1); ORR (Part 2)
- **Case fit:** weak

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Powell/Camidge (2022) — ESMO Open

Reference:PMID 35963179 · **Type:** clinical · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd) — ILD/pneumonitis pooled safety
- **Indication / model:** HER2-positive, HER2-low or HER2-mutant solid tumors (breast 44.3%, gastric 25.6%, lung 17.7%, colorectal 9.3%) (2L+); Median 4 prior lines; median treatment duration 5.8 months. Independent adjudication committee re-reviewed all investigator-reported ILD events.
- **Design:** Pooled safety analysis of nine phase 1-2 monotherapy studies
- **n:** 1150
- **Effect size:** 15.4 % — incidence of adjudicated drug-related ILD/pneumonitis
- **Toxicities:** ILD/pneumonitis (adjudicated drug-related) (n=1150, 15.4%); ILD/pneumonitis (adjudicated drug-related) G1-2 (n=1150); ILD/pneumonitis (adjudicated drug-related) G5 (n=1150, 2.2%)
- **Case fit:** weak
- **Notes:** The risk-factor list includes pre-existing lung comorbidity, which is precisely what the disputed 2026-03 chest CT read would establish if it is correct. Adjudicated onset preceded the investigator-recognised onset in 53.2% of events by a median of 43 days, so baseline imaging ambiguity has a direct cost in this case.

Modi/DESTINY-Breast04 Trial Investigators (2022) — N Engl J Med

Reference:PMID 35665782 · **Type:** clinical · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd) — HER2-low breast (cross-tumor precedent)
- **Indication / model:** HER2-low metastatic breast cancer (IHC 1+, or IHC 2+ with negative ISH) after one or two prior chemotherapy lines (2L+); 88.7% hormone receptor-positive; primary endpoint PFS in the hormone receptor-positive cohort.
- **Design:** Randomized open-label phase 3 (DESTINY-Breast04, NCT03734029)
- **n:** 557
- **Effect size:** 9.9 months — median PFS, all patients
- **Variance:** p=<0.001
- **Toxicities:** any AE G≥3 (n=371, 52.6%); ILD/pneumonitis (adjudicated drug-related) (n=371, 12.1%); ILD/pneumonitis (adjudicated drug-related) G5 (n=371, 0.8%)
- **Case fit:** cross_tumor_only
- **Notes:** The whole basis for treating a HER2-low result as actionable is breast cancer, not urothelial. If the 2026-08 re-stain returns 1+ or 2+/ISH-negative, T-DXd in this patient becomes off-label extrapolation from this trial rather than tumor-agnostic on-label use, which requires IHC 3+.

Meric-Bernstam/Lee (2022) — Lancet Oncol

Reference:PMID 36400106 · **Type:** clinical · **Inclusion:** included

- **Intervention:** zanidatamab (HER2 biparatopic bispecific antibody)
- **Indication / model:** Locally advanced or metastatic HER2-expressing or HER2-amplified solid tumors of any type after all approved therapies (2L+); Part 2 enrolled biliary tract (n=22), colorectal (n=28) and other HER2-expressing/amplified cancers excluding breast and gastroesophageal (n=36).

- **Design:** First-in-human phase 1 dose escalation and expansion (NCT02892123)
- **n:** 132
- **Effect size:** 37.0 % — confirmed ORR (part 2)
- **Variance:** 95% CI 27.0–48.7
- **Toxicities:** diarrhoea (37/86, 43%); infusion-related reaction (29/86, 34%); any treatment-related AE G3 (4/132, 3%); treatment-related death G5 (0/132, 0%)
- **Case fit:** cross_tumor_only
- **Notes:** The non-ADC HER2 route: no cytotoxic payload, so neither the neuropathy nor the ILD constraint applies, at the cost of a lower and less durable response signal outside biliary tract cancer. The registered pan-tumor cohort now bars prior HER2-directed therapy, which current T-DXd triggers.

Kindler/Fennell (2022) — Lancet Oncol

Reference:PMID 35358455 · **Type:** clinical · **Inclusion:** included

- **Intervention:** anetumab ravtansine (mesothelin ADC, DM4 payload)
- **Indication / model:** Unresectable locally advanced or metastatic malignant pleural mesothelioma overexpressing mesothelin, after first-line platinum-pemetrexed (2L+); Prospective screening for mesothelin overexpression, defined as 2+ or 3+ membrane staining in $\geq 30\%$ of viable tumor cells; 248 of 589 screened patients were randomized.
- **Design:** Randomized open-label phase 2 (ARCS-M, NCT02610140)
- **n:** 248
- **Effect size:** 4.3 months — median progression-free survival by blinded central review
- **Variance:** 95% CI 4.1–5.2; $p=0.86$
- **Toxicities:** neutropenia $G \geq 3$ (1/163, 1%); pneumonia $G \geq 3$ (7/163, 4%); dyspnoea $G \geq 3$ (9/163, 6%); any treatment-emergent death G5 (10/163, 6%)
- **Case fit:** cross_tumor_only
- **Notes:** A prospectively mesothelin-selected randomized trial that failed to beat vinorelbine, which is the most important negative result for the mesothelin-ADC concept. Selection on a validated 2+/3+ IHC threshold did not rescue it, so a strong stain alone is not a benefit argument.

Ochsenreither/Garralda (2022) — ESMO Open

Reference:PMID 35397434 · **Type:** clinical · **Inclusion:** included

- **Intervention:** gatipotuzumab (anti-TA-MUC1 antibody) plus an anti-EGFR antibody
- **Indication / model:** Refractory EGFR-positive metastatic solid tumors, then an expansion in colorectal, non-small-cell lung, head and neck and breast cancer (2L+); 20 patients in the primary phase and 32 in the expansion phase. Soluble TA-MUC1 was tracked in serum as a candidate companion biomarker.
- **Design:** Phase 1 dose-finding and expansion (GATTO, NCT03360734)
- **n:** 52
- **Effect size:** 9.6 % — objective response
- **Toxicities:** dose-limiting toxicity (0/52, 0%); related serious adverse event (0/52, 0%); related death G5 (0/52, 0%)
- **Case fit:** cross_tumor_only
- **Notes:** Shows what MUC1 targeting has achieved clinically: safe, with responses confined to colorectal cancer and no urothelial data. It also makes the glycoform point concretely, since the antibody binds the tumor-associated MUC1 epitope rather than MUC1 protein as a multiplex panel would report it. An erratum was

Tan/Toh (2022) — Nat Commun

Reference: [PMID 36307410](#) · Type: clinical · Inclusion: included

- **Intervention:** Ad-sig-hMUC1/ecdCD40L adenoviral vector vaccine
- **Indication / model:** Advanced adenocarcinoma (breast, lung, ovary) (2L+); 21 patients with advanced adenocarcinomas; the trial was not designed to determine clinical efficacy.
- **Design:** Phase 1 dose escalation (NCT02140996)
- **n:** 21
- **Effect size:** 71.0 % — safety and tolerability
- **Toxicities:** injection-site reaction (15/21, 71%); dose-limiting toxicity (0/21, 0%)
- **Case fit:** cross_tumor_only
- **Notes:** A safety-and-immunome study with no efficacy claim, included because it is one of the few completed MUC1-directed clinical trials with a published dataset. It sets the honest ceiling for the MUC1 branch in this case: no response evidence in any tumor type relevant here.

Ferdinandus/Hamacher (2022) — J Nucl Med

Reference: [PMID 34385340](#) · Type: clinical · Inclusion: included

- **Intervention:** 90Y-FAPI-46 FAP-targeted radioligand therapy
- **Indication / model:** Metastatic soft-tissue or bone sarcoma (n=6) and pancreatic cancer (n=3) after exhaustion of approved therapy (2L+); Selection required high FAP expression, defined as 68Ga-FAPI-46 PET SUVmax ≥ 10 in more than 50% of all lesions.
- **Design:** Prospective single-centre case series
- **n:** 9
- **Effect size:** 50.0 % — radiographic disease control
- **Toxicities:** thrombocytopenia $G \geq 3$ (4/9, 44%); anemia $G \geq 3$ (2/9)
- **Case fit:** weak
- **Notes:** Nine patients, no urothelial carcinoma, and selection required intense FAPI PET uptake in most lesions, which is the opposite of what this patient's 2025-09 scan showed for his tumor. Included because it is the published benchmark against which the pending 2026-06 FAP imaging read would have to be judged.

Cabaud/Lopez (2022) — Mol Cancer Ther

Reference: [PMID 35534238](#) · Type: preclinical · Inclusion: included

- **Intervention:** enfortumab vedotin — antigen retention and P-glycoprotein-mediated resistance
- **Indication / model:** breast cancer mouse model treated with an EV-like anti-nectin-4 MMAE ADC (N41mab-vcMMAE) to acquired resistance
- **Design:** Resistance arose through focal ABCB1 amplification and P-glycoprotein overexpression, which pumps MMAE out of the cell while nectin-4 stays on the surface.
- **Effect size:** strong — RNA sequencing of resistant tumors showed ABCB1 upregulation with focal gene amplification and high P-gp protein. Pharmacologic P-gp inhibition restored sensitivity in vitro, and tariquidar plus the ADC produced rapid regression of resistant tumors in mice, whereas tariquidar plus docetaxel was toxic and largely ineffective.

- **Variance:** vs resistant tumors treated with ADC alone; tariquidar plus docetaxel as a comparator combination; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Breast model with an EV-like surrogate rather than clinical EV in urothelial tissue. The load-bearing point for this case is that resistance was a payload-efflux problem, not antigen loss — which argues for keeping nectin-4 on the target list but changing the payload class.

Sailer/Sommermeier (2022) — Cancers (Basel)

Reference: [PMID 35454906](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRAME-specific TCR T cells with a PD1-41BB chimeric costimulatory receptor
- **Indication / model:** PRAME-positive tumor lines with repetitive rechallenge in vitro; PD-L1-positive melanoma xenograft in mice
- **Design:** A high-sensitivity PRAME TCR isolated from non-tolerized repertoires drives killing, while a PD1-41BB switch receptor converts PD-L1 engagement from inhibition into costimulation.
- **Effect size:** strong — Adding PD1-41BB increased IFN-γ secretion and cytotoxicity and prevented exhaustion under repetitive tumor rechallenge, without changing the in-vitro safety profile. A single dose cleared a PD-L1-positive melanoma xenograft that TCR T cells alone could not.
- **Variance:** vs PRAME TCR T cells without the PD1-41BB receptor; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Melanoma xenografts in immunodeficient mice, so the switch-receptor benefit is measured against tumor PD-L1 only, with no host immune contribution. IMA203 as trialed does not carry this receptor.

Kim/Rosenberg (2022) — Cancer Immunol Res

Reference: [PMID 35749374](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** TP53 R175H-directed TCR T cells and p53-reactive tumor-infiltrating lymphocytes
- **Indication / model:** whole-exome screening of metastatic solid tumors with immunologic identification of reactive T cells; TCR-transduced cells tested in vitro and in vivo
- **Design:** TCRs isolated against shared TP53 hotspot neoantigens recognize the mutant peptide only in the context of the matching HLA allele, making the pair — not the mutation alone — the eligibility gate.
- **n:** n=163 patients sequenced; n=78 with TP53 missense mutation; 21 unique T-cell reactivities; 39 TCRs
- **Effect size:** moderate — A library of 39 TCRs covered TP53 mutations shared by 7.3% of patients with solid tumors, recognizing tumor in a mutation- and HLA-specific way. Autologous TIL infused into 12 patients gave only 2 partial responses, with low-frequency, exhausted, poorly persisting p53-reactive cells; one patient given peripheral blood lymphocytes transduced with an HLA-A*02-restricted p53 R175H TCR had a 55% regression lasting 6 months.
- **Variance:** vs autologous TIL without p53 reactivity; HLA-mismatched and wild-type TP53 targets; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** PubMed types this record as an editorial/comment, which is a metadata error — the text is a primary report. The single R175H/HLA-A*02 TCR-transduced patient is the closest human precedent for NT-175 and the transient response is the honest ceiling to quote; it also shows why an engineered TCR product beat unselected

TIL.

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Datopotamab deruxtecan (Dato-DXd) as monotherapy or with volrustomig, rilvegostomig, platinum or fluoropyrimidine
- **Indication / model:** Advanced solid tumors with tumor-specific substudies including urothelial cancer (2L+); TROP2-directed; entry not gated on a TROP2 expression cutoff
- **Design:** Phase 2 modular multi-substudy platform (TROPION-PanTumor03)
- **n:** 454
- **Effect size:** — — Objective response rate; adverse events
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Disitamab vedotin (RC48) alone or with pembrolizumab
- **Indication / model:** Locally advanced or metastatic urothelial cancer expressing HER2 (2L+); HER2 IHC 1+, 2+ or 3+ by central laboratory
- **Design:** Phase 2, multiple cohorts by HER2 tier, line and region
- **n:** 372
- **Effect size:** — — Confirmed objective response rate by blinded independent central review
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Vepugratinib (LY3866288, LOXO-435) alone or with pembrolizumab, enfortumab vedotin or trastuzumab deruxtecan
- **Indication / model:** Urothelial and other solid tumors with a qualifying FGFR3 alteration; cohort B7 requires FGFR3 alteration plus HER2 expression (2L+); Qualifying FGFR3 alteration; cohort B7 additionally requires HER2 expression
- **Design:** Phase 1 dose escalation and expansion (FORAGER-1), multiple urothelial cohorts
- **n:** 677
- **Effect size:** — — Dose-limiting toxicity; objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2022)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** SynKIR-110 (autologous mesothelin KIR-CAR T cells)
- **Indication / model:** Mesothelin-expressing ovarian, primary peritoneal, fallopian tube, cholangiocarcinoma or epithelial mesothelioma (2L+); Mesothelin expression
- **Design:** Phase 1 dose escalation of a KIR-based CAR construct
- **n:** 42
- **Effect size:** — — Safety and tolerability
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Powles/Petrylak (2021) — N Engl J Med

Reference: [PMID 33577729](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** enfortumab vedotin (Nectin-4 ADC, MMAE payload)
- **Indication / model:** Locally advanced or metastatic urothelial carcinoma after platinum-containing chemotherapy and progression on PD-1/PD-L1 inhibition (2L+); Prior platinum was required; this patient never received platinum, so the trial population is one line to the left of his course.
- **Design:** Global open-label randomized phase 3 (EV-301, NCT03474107)
- **n:** 608
- **Effect size:** 12.88 months — median overall survival
- **Variance:** $p=0.001$
- **Toxicities:** any treatment-related AE ($n=296$, 93.9%); any treatment-related AE $G \geq 3$ ($n=296$, 51.4%)
- **Case fit:** partial
- **Notes:** Establishes Nectin-4 as a survival-relevant target in urothelial carcinoma, in a post-platinum population this patient does not belong to. Per-term neuropathy and skin-toxicity rates sit in the paper's supplementary tables rather than the abstract; the abstract-level union rates are what is captured here.

Tagawa/Grivas (2021) — J Clin Oncol

Reference: [PMID 33929895](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** sacituzumab govitecan (TROP2 ADC, SN-38 payload)
- **Indication / model:** Locally advanced or metastatic urothelial carcinoma after platinum-based chemotherapy and a checkpoint inhibitor (2L+); Median three prior therapies; 66.4% visceral metastases. Entry required prior platinum, which this patient has not had.
- **Design:** Single-arm cohort 1 of a multicohort phase 2 registrational study (TROPY-U-01, NCT03547973)
- **n:** 113
- **Effect size:** 27.0 % — ORR by central review
- **Variance:** 95% CI 19.5–36.6
- **Toxicities:** neutropenia $G \geq 3$ ($n=113$, 35%); febrile neutropenia $G \geq 3$ ($n=113$, 10%); leukopenia $G \geq 3$ ($n=113$, 18%); anemia $G \geq 3$ ($n=113$, 14%); diarrhea $G \geq 3$ ($n=113$, 10%); treatment-related AE leading to discontinuation ($n=113$, 6%)
- **Case fit:** partial
- **Notes:** The cohort that produced the US accelerated approval later withdrawn in October 2024 after TROPiCS-04. Entry required prior platinum, so this patient would not have qualified, and the diarrhoea signal

interacts with his current gastrointestinal intolerance.

Banerjee/Bliss (2021) — Int J Gynecol Cancer

Reference:PMID 34518240 · **Type:** clinical · **Inclusion:** excluded — Trial-in-progress protocol paper with no results reported.

- **Intervention:** ceralasertib plus olaparib in ARID1A-loss gynecological cancers (ATARI)
- **Notes:** Excluded: Trial-in-progress protocol paper with no results reported.

Chu/Chou (2021) — Clin Cancer Res

Reference:PMID 34108177 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** enfortumab vedotin — NECTIN4 expression as the sensitivity determinant
- **Indication / model:** seven muscle-invasive bladder cancer transcriptomic cohorts plus luminal and basal bladder cancer cell lines with NECTIN4 overexpression or knockdown
- **Design:** NECTIN4 sits downstream of luminal differentiation (GATA3, FOXA1, PPARG), and surface density sets how much MMAE reaches the cell.
- **n:** n=1,915 specimens across cohorts
- **Effect size:** strong — NECTIN4 expression was heterogeneous across consensus molecular subtypes and enriched in luminal tumors. Knockdown conferred EV resistance and overexpression conferred sensitivity in both luminal and basal lines, making expression both necessary and sufficient.
- **Variance:** vs isogenic parental lines; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Cell-line manipulation rather than post-EV patient tissue, and subtype was not assigned for this patient. It is the reason a contemporary nectin-4 stain, not the 2023 result, is the decision-relevant test.

Hsiue/Zhou (2021) — Science

Reference:PMID 33649166 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** H2-scDb (TP53 R175H/HLA-A*02:01-directed bispecific single-chain diabody)
- **Indication / model:** human cancer cell lines presenting p53 R175H on HLA-A*02:01, with T cells, and mouse xenograft models
- **Design:** An antibody selected against the HMTEVVRHC peptide bound to HLA-A*02:01 is built into a bispecific diabody that also engages CD3, redirecting T cells against a mutant-p53-derived neoantigen.
- **Effect size:** strong — The antibody bound the mutant peptide-HLA complex with specificity resolved structurally, discriminating R175H from wild-type sequence. Despite very low peptide-HLA copy number on the cancer cell surface, the bispecific activated T cells to lyse antigen-presenting cancer cells in vitro and in mice.
- **Variance:** vs wild-type TP53 and HLA-A2-negative cell lines; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Cell lines and xenografts, no urothelial model, and no clinical publication exists for CLSP-1025 — this paper and its successors are the entire evidence base for that trial. This patient's R175H was seen in plasma ctDNA at 0.09% VAF, so whether the current tumor clone even carries it is unestablished.

Yamada/Kono (2021) — Gastric Cancer

Reference: [PMID 32506298](#) · Type: preclinical · Inclusion: included

- **Intervention:** EZH2 inhibition in ARID1A-deficient cancer
- **Indication / model:** ARID1A-deficient and ARID1A-wild-type gastric cancer cell lines, including ARID1A knockdown; cell viability and colony formation
- **Design:** Loss of the SWI/SNF subunit ARID1A leaves the tumor dependent on PRC2/EZH2-driven repression, with PI3K/AKT signaling as the downstream survival axis.
- **Effect size:** moderate — EZH2 inhibitors reduced viability of ARID1A-deficient cells dose-dependently and selectively relative to ARID1A-wild-type cells. ARID1A knockdown raised phosphorylated AKT, and EZH2 downregulation normalized PI3K/AKT signaling in the deficient cells.
- **Variance:** vs ARID1A-wild-type parental lines; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** In vitro only, no xenograft arm, gastric rather than urothelial. No EZH2-inhibitor trial appears on this case's list, so it stays as background for the synthetic-lethality axis rather than a route to an option.

—/— (2021)

Reference: — · Type: trial · Inclusion: included

- **Intervention:** [¹⁷⁷Lu]Lu-FAP-2286 radioligand therapy, gated by [⁶⁸Ga]Ga-FAP-2286 PET
- **Indication / model:** Advanced solid tumors with ⁶⁸Ga-FAP-2286 uptake (2L+); FAP expression demonstrated by positive [⁶⁸Ga]Ga-FAP-2286 uptake
- **Design:** Phase 1/2 theranostic pair, dose escalation then tumor-specific groups
- **n:** 222
- **Effect size:** — — DLT and RP2D (phase 1); objective response rate (phase 2)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Marabelle/Bang (2020) — Lancet Oncol

Reference: [PMID 32919526](#) · Type: clinical · Inclusion: included

- **Intervention:** pembrolizumab in tissue TMB-high solid tumors
- **Indication / model:** Previously treated advanced anal, biliary, cervical, endometrial, mesothelioma, neuroendocrine, salivary, small-cell lung, thyroid and vulvar cancers (2L+); 790 patients evaluable for TMB; 102 (13%) had tTMB ≥10 mut/Mb measured on FoundationOne CDx. Urothelial carcinoma was not one of the ten enrolled cohorts.
- **Design:** Prospective biomarker analysis within a multi-cohort phase 2 study (KEYNOTE-158, NCT02628067)
- **n:** 790
- **Effect size:** 29.0 % — objective response rate in the tTMB-high group
- **Variance:** 95% CI 21.0–39.0
- **Toxicities:** any treatment-related AE G≥3 (16/105, 15%); treatment-related serious adverse event (11/105, 10%); colitis G≥3 (2/105, 1.9%); pneumonia G5 (1/105, 1%)
- **Case fit:** weak

- **Notes:** The registration evidence behind the tumor-agnostic TMB-high pembrolizumab indication, and the reason the ≥ 10 mut/Mb cutoff exists. Two problems apply here: the threshold was validated on FoundationOne CDx rather than the institutional 468-gene panel that produced this patient's 12.4 mut/Mb, and he has already progressed on prolonged pembrolizumab.

Yap/de Bono (2020) — J Clin Oncol

Reference:PMID 32568634 · **Type:** clinical · **Inclusion:** included

- **Intervention:** berzosertib (M6620/VX-970, ATR inhibitor) alone and with carboplatin
- **Indication / model:** Advanced solid tumors, unselected for ARID1A (2L+); 17 patients received monotherapy and 23 the carboplatin combination. Molecular annotation was retrospective rather than an entry criterion.
- **Design:** First-in-class phase 1 dose escalation
- **n:** 40
- **Effect size:** one durable complete response (ATM loss plus ARID1A mutation), one partial response, 15 with stable disease — — antitumor activity and recommended phase 2 dose
- **Toxicities:** hematologic toxicity (combination with carboplatin) $G \geq 3$ (n=23)
- **Case fit:** weak
- **Notes:** The single durable complete response in this first-in-class study occurred in a tumor carrying both ATM loss and an ARID1A mutation, which is the anecdote the ARID1A-ATR hypothesis is usually built on. One patient in an unselected phase 1 is not a benefit estimate, and ARID1A alone was never the selection criterion.

Hassan/Bendell (2020) — J Clin Oncol

Reference:PMID 32213105 · **Type:** clinical · **Inclusion:** excluded — Superseded for the mesothelin-ADC question by the randomized ARCS-M readout, which selected on validated mesothelin IHC and was negative.

- **Intervention:** anetumab ravtansine first-in-human phase 1
- **Notes:** Excluded: Superseded for the mesothelin-ADC question by the randomized ARCS-M readout, which selected on validated mesothelin IHC and was negative.

Shigeta/Oya (2020) — Cancer Sci

Reference:PMID 32677159 · **Type:** preclinical · **Inclusion:** excluded — Retracted (PubMed publication type 'Retracted Publication'); its MUC1-C/MDR1 findings in urothelial carcinoma cannot be relied on.

- **Intervention:** MUC1-C inhibition in cisplatin-resistant urothelial carcinoma
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Retracted (PubMed publication type 'Retracted Publication'); its MUC1-C/MDR1 findings in urothelial carcinoma cannot be relied on.

Malekzadeh/Deniger (2020) — Clin Cancer Res

Reference:PMID 31996390 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** p53 neoantigen-reactive T cells recovered from peripheral blood
- **Indication / model:** sorted antigen-experienced peripheral blood lymphocytes from patients with TP53-mutant tumors, stimulated in vitro against p53 neoantigens; tumor cell line recognition assays

- **Design:** Antigen-experienced circulating T cells mirror the intratumoral p53-neoantigen repertoire, so blood can substitute for tumor as the starting material for identifying reactive TCRs.
- **Effect size:** moderate — All five patients with p53-reactive TIL had matching reactivity in blood, and the four without TIL reactivity had none in blood either. One culture was 78% reactive against p53 R175H on HLA-A*02:01, and TP53-mutation-specific T cells recognized tumor lines carrying the right HLA and mutation, confirming natural processing and presentation.
- **Variance:** vs patients without TIL reactivity to mutated TP53; wild-type p53 and HLA-mismatched targets; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Small patient numbers and no clinical outcome. Directly useful given this case's tissue scarcity: banked PBMCs and apheresis could be screened for R175H/HLA-A*02:01 reactivity without spending the one fresh-frozen tumor vial.

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** CAdVEC oncolytic adenovirus intratumorally plus autologous HER2-specific CAR virus-specific T cells
- **Indication / model:** Advanced refractory HER2-positive solid tumors, bladder cancer among the named histologies (2L+); HER2 IHC $\geq 2+$ (CB11 or 4B5), greater than weak-to-moderate intensity in $>10\%$ of tumor cells
- **Design:** Phase 1 BOIN dose escalation, basket across HER2-positive solid tumors
- **n:** 45
- **Effect size:** — — Dose-limiting toxicity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Brenetafusp (IMC-F106C) alone or with checkpoint inhibitors, chemotherapy or targeted therapy
- **Indication / model:** PRAME-positive advanced solid tumors in HLA-A02:01-positive patients (2L+); HLA-A 02:01 positive and PRAME-positive tumor
- **Design:** Phase 1/2 dose escalation and histology-specific expansion
- **n:** 410
- **Effect size:** — — Dose-limiting toxicity; adverse events
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** ART0380 (ATR inhibitor) alone or with gemcitabine or irinotecan
- **Indication / model:** Advanced or metastatic solid tumors, with DNA-damage-response-selected expansion cohorts (2L+); Dose escalation unselected; expansion cohorts require ATM loss or BRCA/HRD positivity
- **Design:** Phase 1/2 dose escalation (parts A1-A3) and molecularly selected expansion (parts B1-B2)
- **n:** 542
- **Effect size:** — — Recommended dose; objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Sharma/Rosenberg (2019) — J Clin Oncol

Reference: PMID 31100038 · **Type:** clinical · **Inclusion:** included

- **Intervention:** nivolumab 1 mg/kg plus ipilimumab 3 mg/kg
- **Indication / model:** Platinum-pretreated locally advanced or metastatic urothelial carcinoma (2L+); Three cohorts compared: nivolumab 3 mg/kg alone (n=78), nivolumab 3 plus ipilimumab 1 (n=104), nivolumab 1 plus ipilimumab 3 (n=92). Patients were checkpoint-naïve at entry.
- **Design:** Open-label multicohort phase 1/2 (CheckMate 032, NCT01928394)
- **n:** 274
- **Effect size:** 38.0 % — investigator-assessed ORR, nivolumab 1 plus ipilimumab 3 cohort
- **Toxicities:** any treatment-related AE G3-4 (36/92, 39.1%); any treatment-related AE G3-4 (32/104, 30.8%); any treatment-related AE G3-4 (21/78, 26.9%); pneumonitis G5 (2/182, 1.1%)
- **Case fit:** partial
- **Notes:** Establishes the dose dependence of the CTLA-4 contribution in urothelial carcinoma: response climbed with ipilimumab dose and so did grade 3-4 toxicity. Patients here were checkpoint-naïve, so it estimates the value of adding ipilimumab up front rather than after PD-1 failure. An erratum was published in J Clin Oncol 2019 (10.1200/JCO.19.01700).

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** IMA203 / IMA203CD8 PRAME-directed TCR-T, alone or with nivolumab
- **Indication / model:** Recurrent or refractory PRAME-positive solid tumors in HLA-A02:01-positive patients (2L+); HLA-A02:01 positive AND PRAME expression by IMADetect RT-qPCR
- **Design:** Phase 1/2 dose escalation with disease-specific extension cohorts including a rare-cancer cohort
- **n:** 375
- **Effect size:** — — MTD and recommended dose; tumor response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2019)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Cabozantinib with nivolumab and ipilimumab

- **Indication / model:** Rare genitourinary tumors, including plasmacytoid and other variant-histology urothelial carcinoma (2L+); Variant histology by central pathology; no molecular gate
- **Design:** Phase 2 multi-cohort by rare GU histology (ICONIC)
- **n:** 314
- **Effect size:** — — Objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2018)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Sacituzumab govitecan alone or with pembrolizumab, zimberelimab, avelumab, cisplatin, domvanalimab or enfortumab vedotin
- **Indication / model:** Metastatic urothelial cancer, including a platinum-ineligible cohort after first-line checkpoint therapy (2L+); TROP2-directed; no expression cutoff for entry
- **Design:** Phase 2 multi-cohort (TROPY-U-01); cohort 2 is the platinum-ineligible, post-checkpoint cohort
- **n:** 827
- **Effect size:** — — Objective response rate
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2018)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Individualized neoantigen vaccine with durvalumab, tremelimumab and chemotherapy
- **Indication / model:** Metastatic triple-negative breast cancer (2L+); Personalized neoepitopes predicted from tumor sequencing
- **Design:** Randomized phase 2 adding a personalized vaccine to dual checkpoint blockade and chemotherapy
- **n:** 86
- **Effect size:** — — Progression-free survival
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Obara/Fujioka (2017) — Ann Oncol

Reference: [PMID 27998971](#) · **Type:** clinical · **Inclusion:** excluded — Shared tumor-antigen peptide vaccine rather than a personalized neoantigen construct; the mechanism does not match the neoantigen axis in this profile.

- **Intervention:** S-288310 shared-antigen peptide vaccine in urothelial carcinoma
- **Notes:** Excluded: Shared tumor-antigen peptide vaccine rather than a personalized neoantigen construct; the mechanism does not match the neoantigen axis in this profile.

Chang/Scheinberg (2017) — J Clin Invest

Reference:PMID 28628042 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** PRAME peptide/HLA-A*02:01-directed TCR-mimic antibody (Pr20)
- **Indication / model:** PRAME-positive/HLA-A2-positive human cancer and leukemia lines; mouse xenograft models of human leukemia
- **Design:** An antibody shaped like a TCR binds the PRAME300-309 (ALYVDSLFFL) peptide presented on HLA-A*02:01, converting an intracellular antigen into a surface target.
- **Effect size:** moderate — Pr20 bound PRAME-positive HLA-A2-positive cancers, and the afucosylated form drove ADCC and was effective in leukemia xenografts. Binding rose sharply after IFN- γ exposure through induction of the immunoproteasome subunit $\beta 5i$, which spares the ALY epitope from destructive internal cleavage.
- **Variance:** vs PRAME-negative and HLA-A2-negative lines; unmodified IgG1 versus afucosylated Fc; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Leukemia xenografts, not solid tumor, and Pr20 is not brenetafusp. Two transferable points: the epitope is presented at very low copy number, and its density is IFN- γ -dependent — so a PRAME-directed agent's target may vary with the inflammatory state of the lesion, not only with PRAME transcript level.

—/— (2017)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** N-803 (IL-15 superagonist) with pembrolizumab, nivolumab or atezolizumab, with or without PD-L1 t-haNK cells
- **Indication / model:** Checkpoint-inhibitor-experienced solid tumors including urothelial carcinoma (2L+); Prior checkpoint exposure; no molecular gate
- **Design:** Phase 2 multi-cohort (QUILT-3.055) of checkpoint rechallenge with IL-15 superagonist
- **n:** 40
- **Effect size:** — — Objective response rate
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Ogitani/Agatsuma (2016) — Cancer Sci

Reference:PMID 27166974 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** trastuzumab deruxtecan (T-DXd) — DXd bystander mechanism
- **Indication / model:** HER2-positive/HER2-negative tumor cell cocultures (KPL-4 + MDA-MB-468) and mixed-cell mouse xenografts (NCI-N87 + MDA-MB-468-Luc)
- **Design:** The released DXd payload crosses membranes into neighboring cells, so killing is not confined to cells that internalize the ADC.
- **Effect size:** strong — DS-8201a killed HER2-negative cells cocultured with HER2-positive cells, while T-DM1 and a low-permeability DXd conjugate did not. In mixed NCI-N87/MDA-MB-468-Luc xenografts DS-8201a suppressed the HER2-negative luciferase signal. HER2-negative tumors implanted on the contralateral flank were unaffected, placing the effect at the range of a few cell diameters.
- **Variance:** vs T-DM1 and an anti-HER2 ADC carrying a low-permeability DXd conjugate; translatability: med

- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Breast and gastric lines only, no urothelial model. The bystander range is local, so it rescues heterogeneous or antigen-low tumor rather than antigen-null tumor — a distinction that matters if the current node has genuinely lost HER2.

Williamson/Lord (2016) — Nat Commun

Reference: [PMID 27958275](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** ATR inhibition as synthetic lethality in ARID1A-deficient tumors
- **Indication / model:** isogenic and naturally ARID1A-mutant tumor cell lines in vitro, with in-vivo confirmation
- **Design:** ARID1A loss produces topoisomerase 2A and cell-cycle defects that leave the cell dependent on ATR checkpoint activity; removing ATR forces premature mitotic entry.
- **Effect size:** strong — ARID1A defects sensitized tumor cells to ATR inhibitors in vitro and in vivo. In ARID1A-mutant cells, ATR inhibition triggered premature mitotic entry, genomic instability and apoptosis.
- **Variance:** vs ARID1A-proficient isogenic cells; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** No urothelial model, and the case's ARID1A alteration is reported without zygosity or protein-loss confirmation — the biomarker in these experiments is ARID1A deficiency, not the presence of a mutation call. This is the original rationale behind the ATR-inhibitor trials on the list (ART0380, tuvusertib) and remains hypothesis-grade for this patient.

—/— (2016)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Trastuzumab plus pertuzumab (HER2 cohort) and other FDA-approved targeted agents matched to genomic findings
- **Indication / model:** Advanced solid tumors, multiple myeloma or B-cell non-Hodgkin lymphoma with a targetable genomic variant (2L+); ERBB2 amplification or HER2 overexpression for the HER2 cohort; genomic report required
- **Design:** Non-randomized phase 2 pragmatic basket, drug assignment by genomic match
- **n:** 4200
- **Effect size:** — — Objective response or stable disease at 16 weeks
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Spranger/Gajewski (2014) — J Immunother Cancer

Reference: [PMID 24829760](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** CTLA-4 plus PD-1/PD-L1 blockade — intratumoral mechanism of the doublet
- **Indication / model:** B16.SIY murine melanoma
- **Design:** The doublet acts on T cells already inside the tumor, restoring IL-2 production and proliferation, rather than by priming new T cells in draining lymph nodes.

- **Effect size:** strong — All three doublets improved tumor control over single agents, with complete rejection in many mice and no vaccine or adoptive transfer. Priming in draining nodes rose only minimally, while tumor-infiltrating CD8 T cells regained IL-2 production and proliferation by day 3; blocking new T-cell trafficking with FTY720 preserved most of the effect.
- **Variance:** vs single-agent anti-CTLA-4, anti-PD-L1 or IDO inhibition; FTY720-treated mice to block lymphocyte egress; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** A single immunogenic murine melanoma line, and the mice were checkpoint-naïve. This patient progressed on pembrolizumab maintenance, so the relevant question is whether an exhausted intratumoral pool still responds to CTLA-4 addition — which this model does not test.

—/— (2010)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Neoantigen-reactive tumor-infiltrating lymphocytes with aldesleukin, after lymphodepletion, with or without pembrolizumab
- **Indication / model:** Metastatic genitourinary, gastrointestinal, hepatobiliary, breast, gynecologic or endocrine cancers (2L+); Tumor-reactive T cells against mutation-derived neoantigens identified from resected tumor
- **Design:** Phase 2 adoptive cell transfer, NCI Surgery Branch
- **n:** 332
- **Effect size:** — — Objective tumor response
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)